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Demyelinating Diseases

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No Disclosures

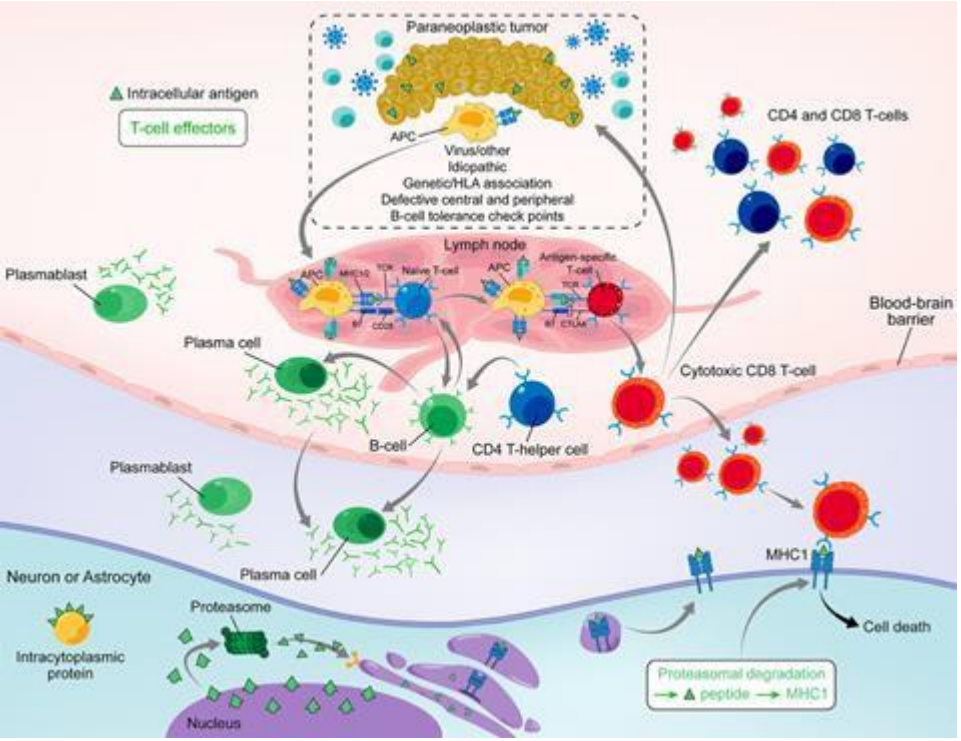
- Describe the pathologic mechanisms of demyelinating disease in the CNS & PNS
- Compare the clinical features & diagnostic approaches of MS (Multiple Sclerosis) & GBS (Guillain Barre Syndrome)
- Explain the immunopathogenesis & diagnostic criteria of MS, including relevant imaging & laboratory features
- Touch on differential diagnoses i.e., NMOSD, MOGAD, CIDP

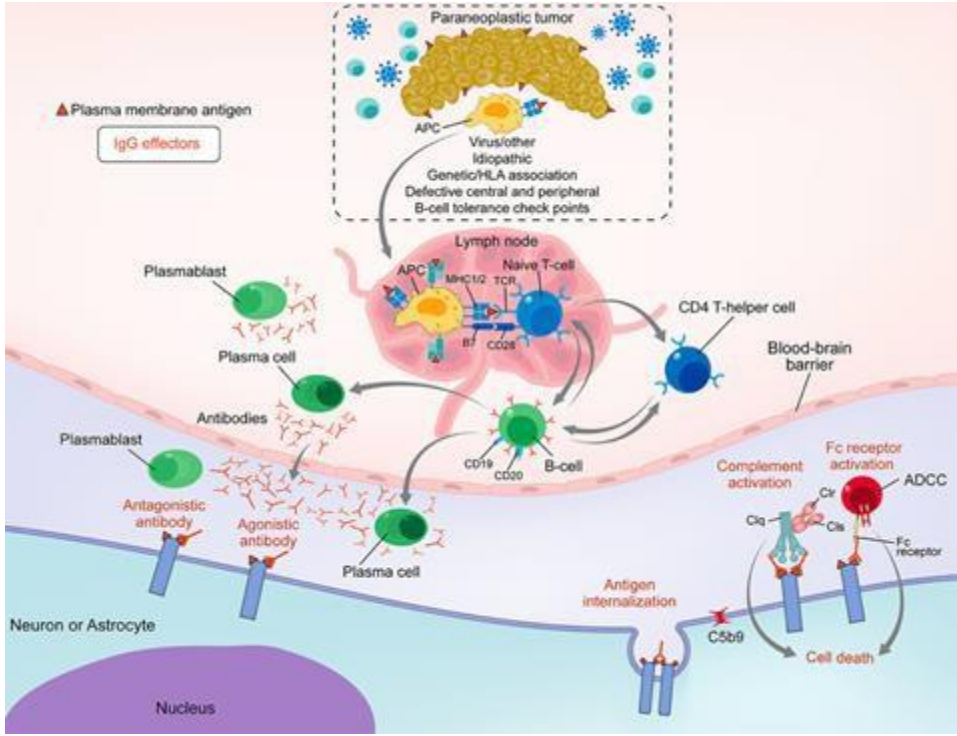
Objectives

Immunopathologic mechanisms of paraneoplastic & idiopathic neural antibodies targeting neural

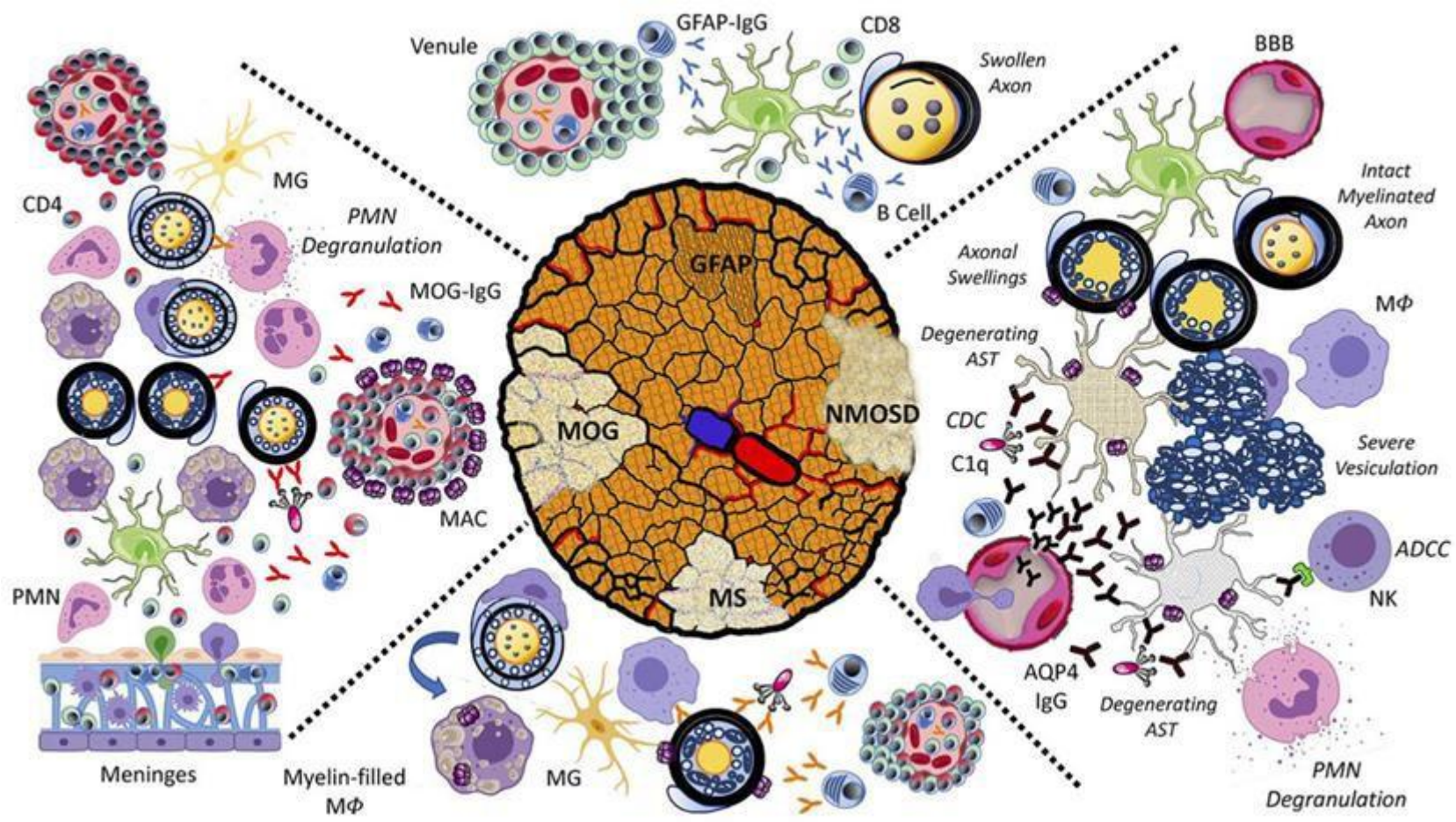
intracellular (nuclear or cytoplasmic) antigens

cell plasma membrane antigens

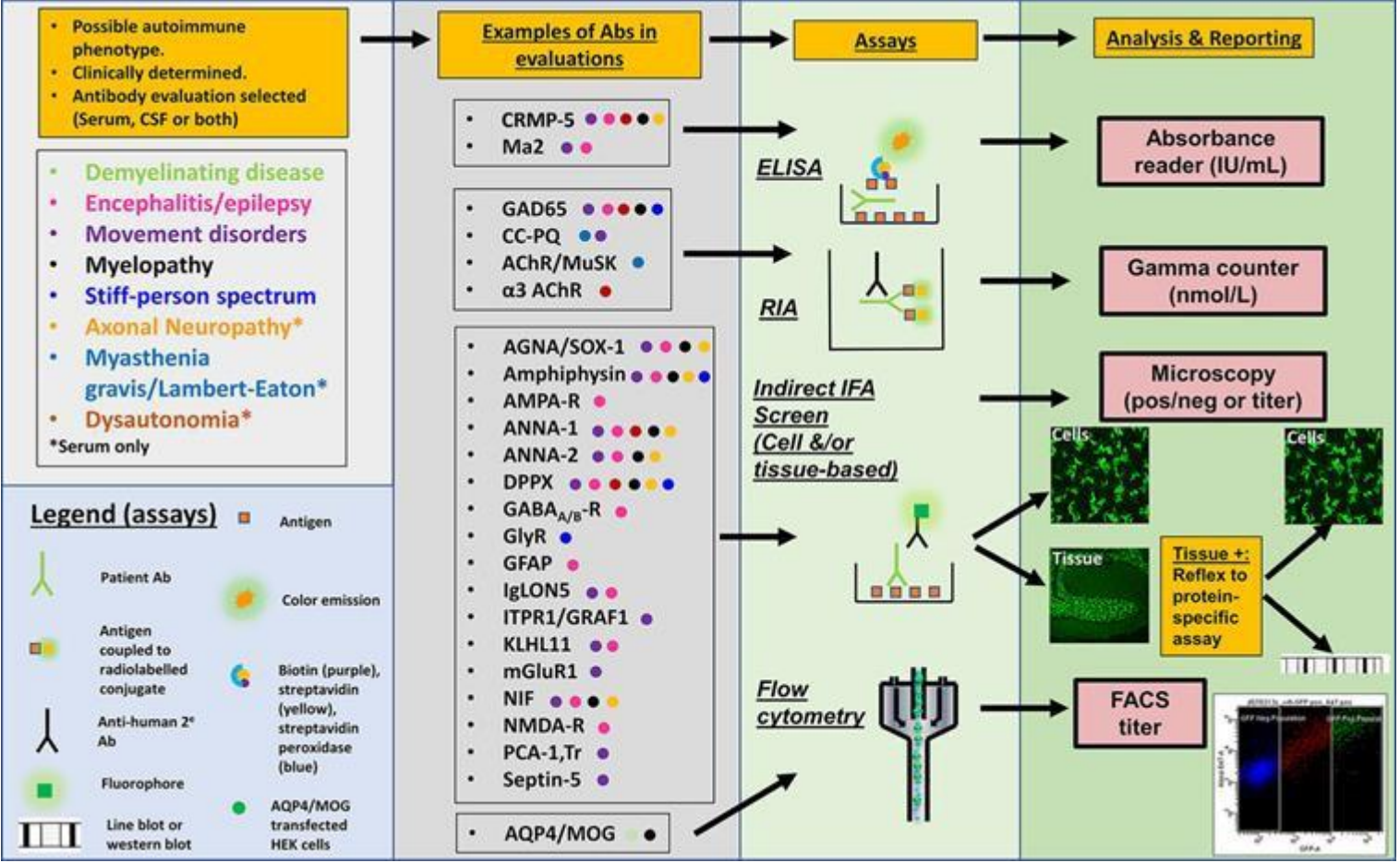




CTLA4 = Cytotoxic T-Lymphocyte Associated Protein 4; ADCC = Antibody-Dependent Cellular Cytotoxicity; HLA = Human Leukocyte Antigen; MHC = Major Histocompatibility Complex; TCR = T Cell Receptor



Immune Pathophysiology

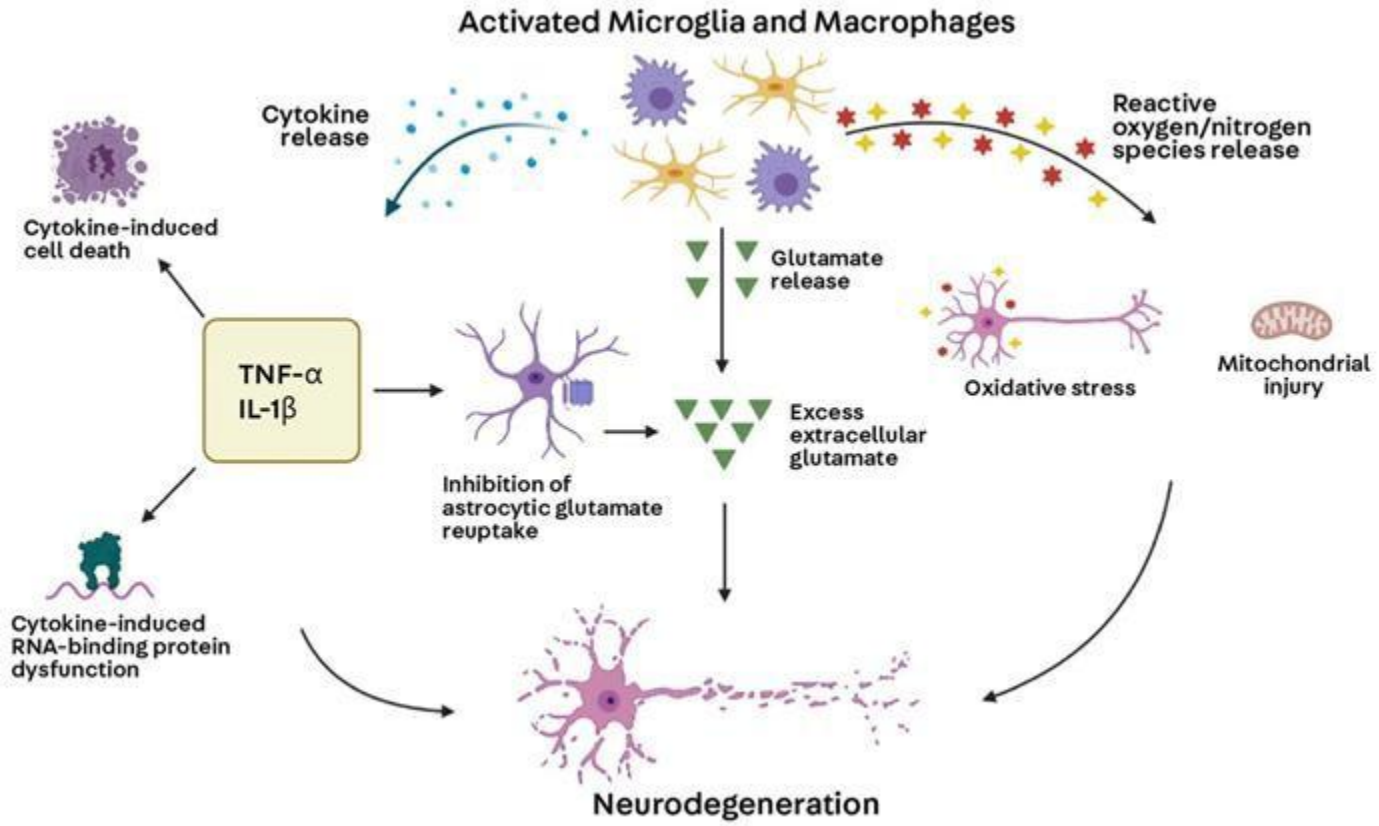


AChR = acetylcholine receptor
AGNA = antiglial nuclear antibody
AMPA =α-amino-3-hydroxy-5-methyl-4-isoxazole propionic acid
ANNA-2 = antineuronal nuclear antibody type 2
AQP4 = aquaporin-4
CC-PQ = P/Q-type voltage-gated calcium channel
CRMP-5 = collapsin response mediator protein-5
DPPX = dipeptidyl-peptidase–like protein 6
ELISA = enzyme-linked immunosorbent assay
FACS = fluorescence-activated cell sorting assay
GABA = γ-aminobutyric acid
GAD65 = glutamic acid decarboxylase 65
GlyR = glycine receptor
GRAF 1 = GTPase regulator associated with focal adhesion kinase-1
HEK = human embryonic kidney
IgLON5 = Ig-like cell adhesion molecule 5
ITPR1 = inositol 1,4,5-trisphosphate receptor type 1
KLHL11 = Kelch-like protein 11
mGluR1 = metabotropic glutamate receptor 1
MOG = myelin oligodendrocyte glycoprotein
MuSK = muscle-specific tyrosine kinase
NIF = neuronal intermediate filament
NMDA-R = *N*-methyl-D-aspartate receptor
PCA-1, Tr = Purkinje cell cytoplasmic antibody Tr
RIA = radioimmunoprecipitation assay.

Neuroimmunology Labs

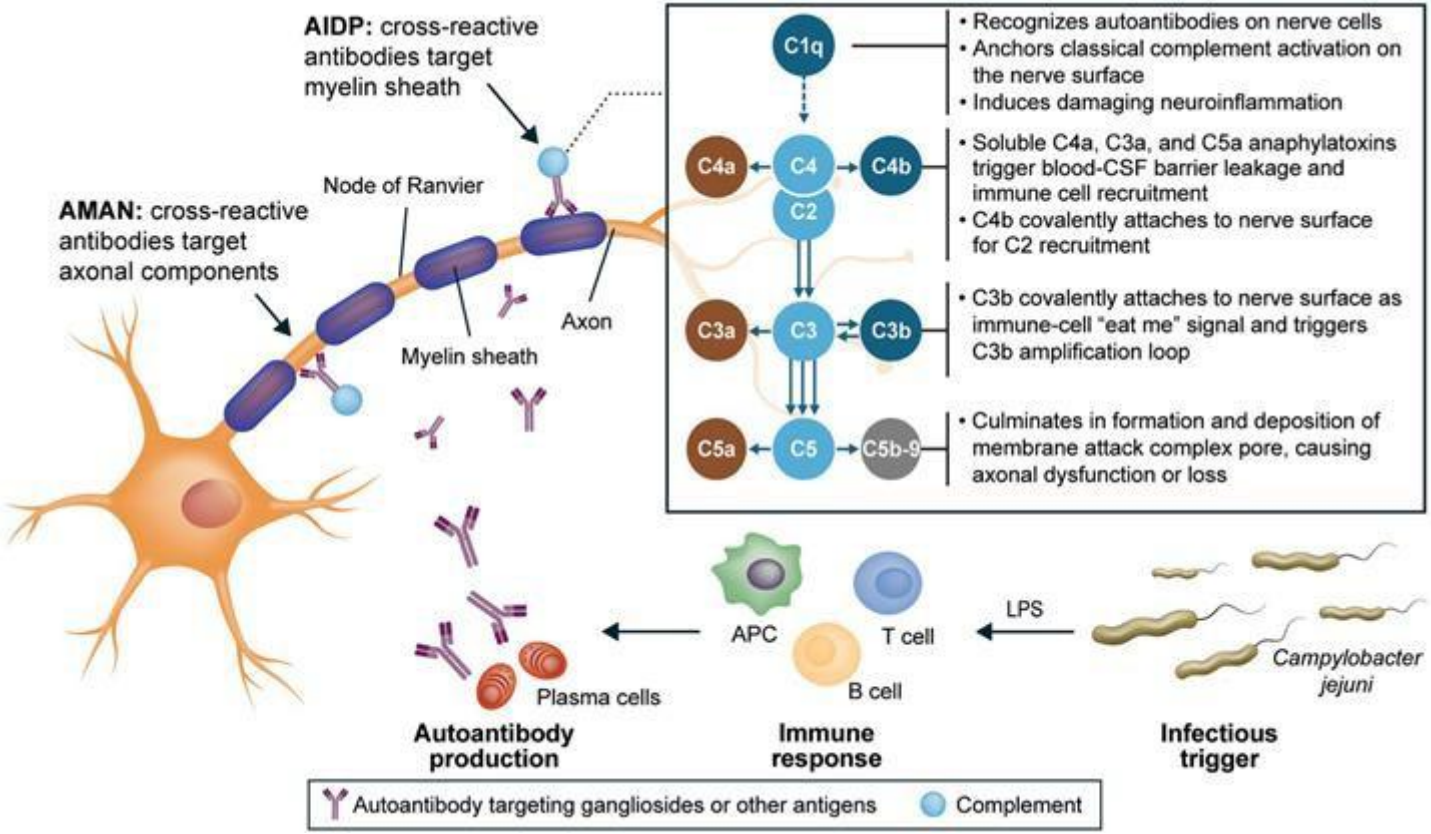
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Autoimmune demyelination T cell, B cell, macrophage mediated inflammation targeting oligodendrocytes in CNS

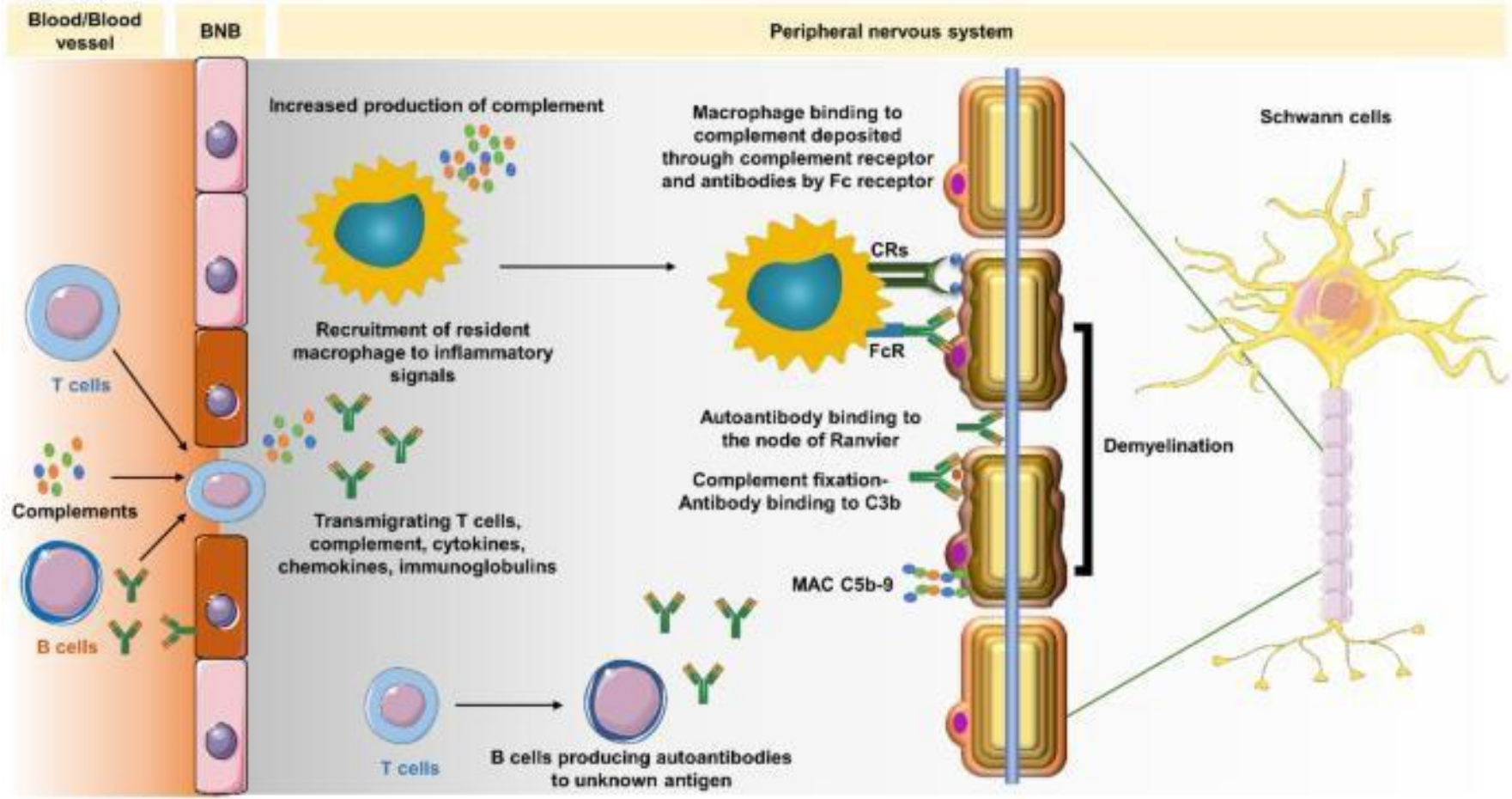


MS Pathophysiology

Autoimmune mediated endoneurial inflammation segmental demyelination PNS Schwann cells,
often triggered by infections (e.g., C. jejuni dysentery or respiratory) via molecular mimicry bacterial lipo-oligosaccarides & gangliosides

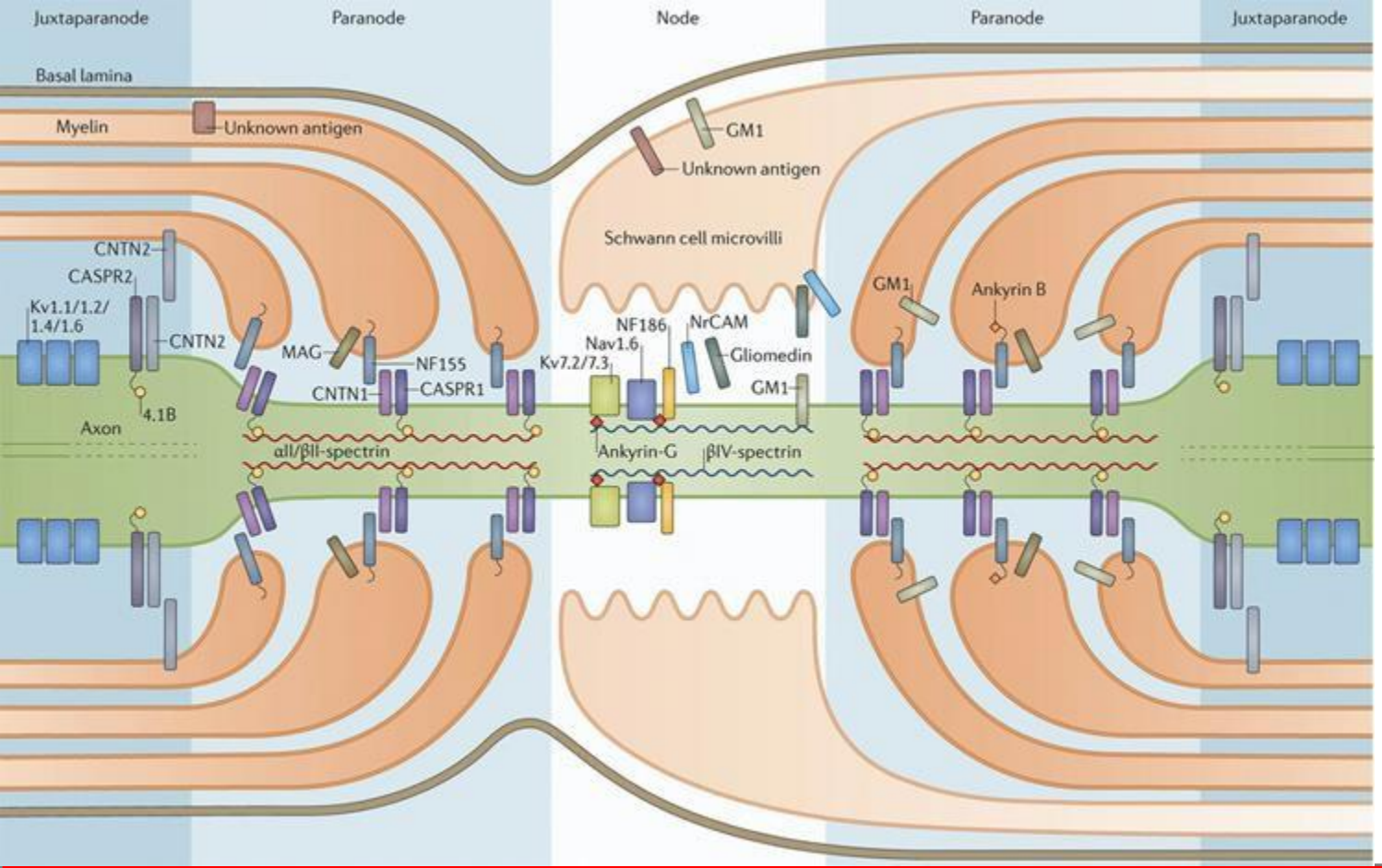


GBS Pathophysiology

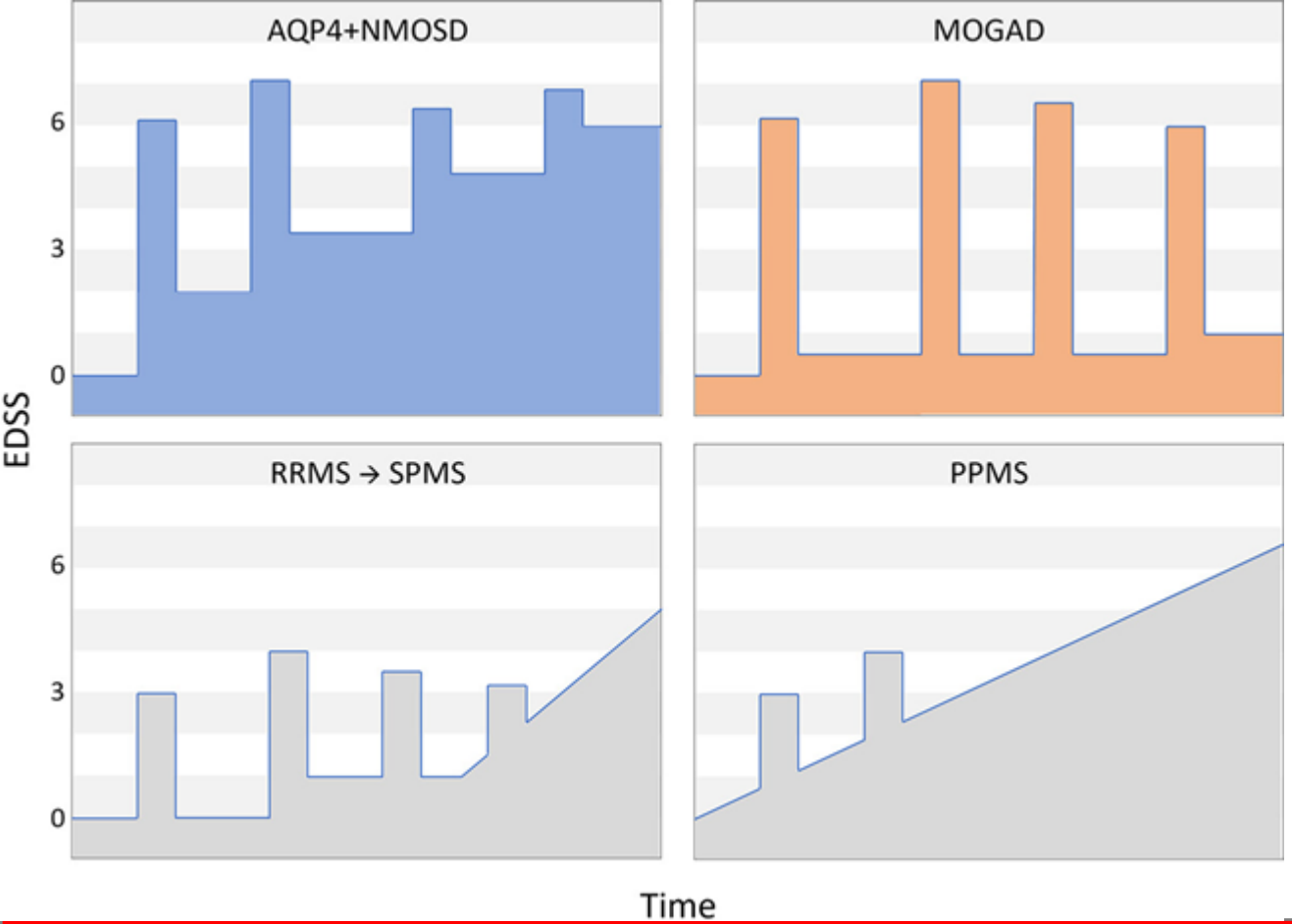


(CIDP Pathophysiology)

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(Nodo-Paranodopathies)



Disease courses & accumulated disability patterns / Expanded Disability Status Scale (EDSS)

Aquaporin-4 antibody–positive Neuromyelitis Optica Spectrum Disorder (AQP4-NMOSD)

Myelin Oligodendrocyte Glycoprotein antibody–Associated Disease (MOGAD)

RRMS = Relapsing-Remitting Multiple Sclerosis

SPMS = Secondary Progressive Multiple Sclerosis

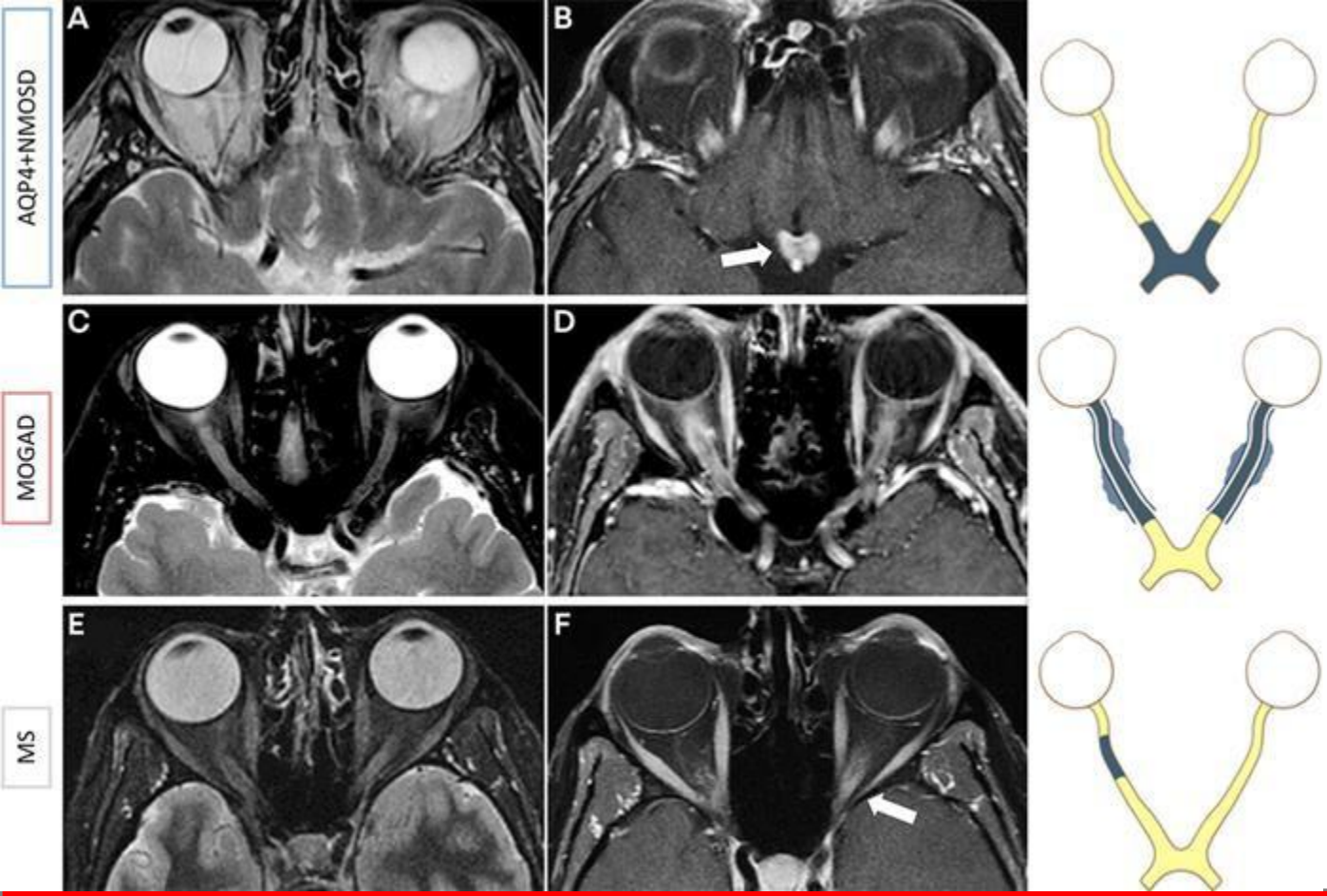
PPMS = Primary Progressive Multiple Sclerosis

Disease Courses & Disability

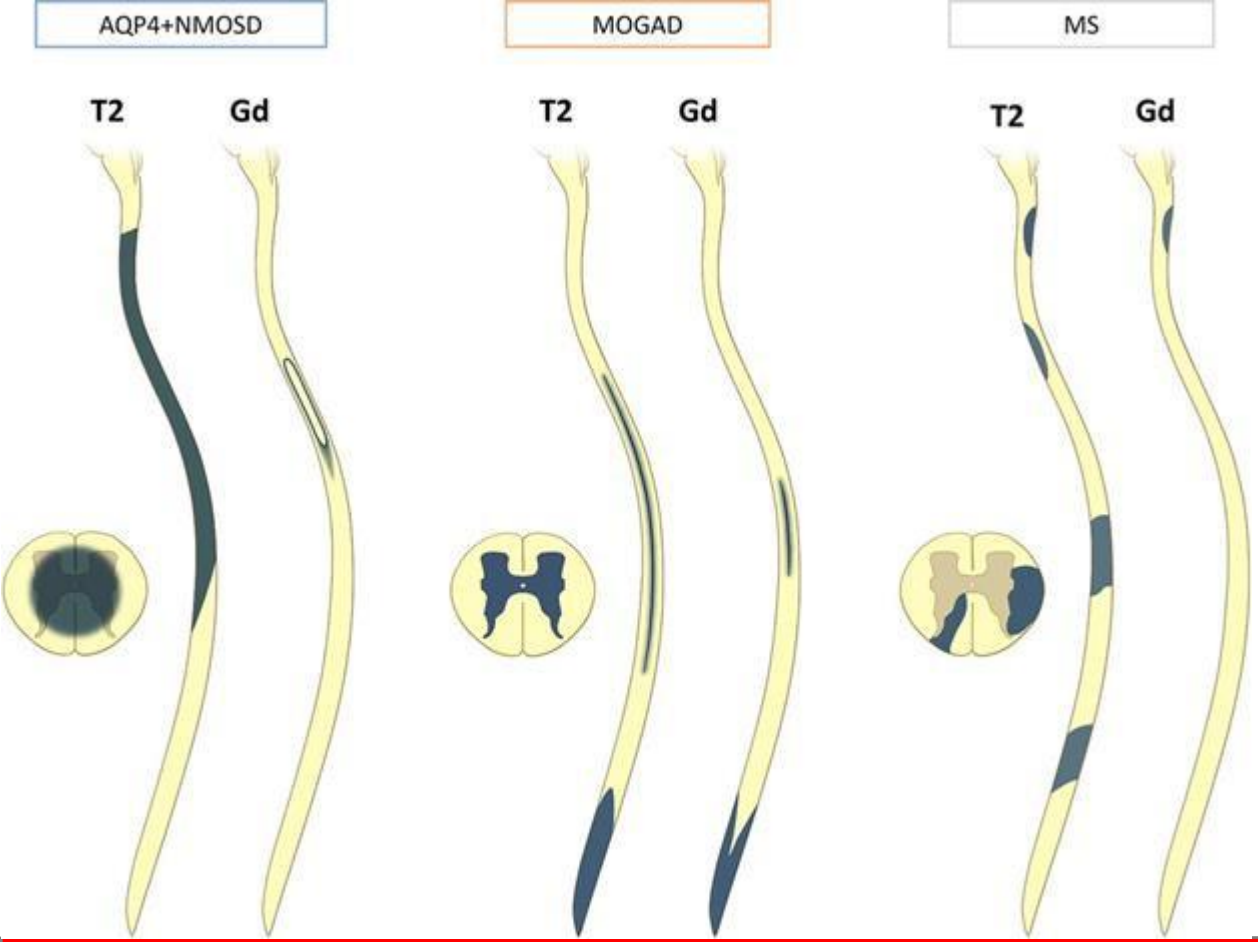
	AQP4-NMOSD	MOGAD	Multiple sclerosis
Demographics			
Most common age at onset (years)	30-50	0-40	20-40
Onset age < 18 years	Rare	Common	Infrequent
Sex (F:M)	9:1	1:1	3:1
Clinical features			
Antecedent infection or immunization	Infrequent	Common	Infrequent
Disease course	Generally relapsing (>90%); a progressive course is rare	Relapsing (50%) or monophasic (50%); a progressive course is rare	Relapsing, secondary progressive or primary progressive
Optic neuritis	Common (unilateral or bilateral)	Common (often bilateral)	Common (often unilateral)
Myelitis	Common	Common	Common
Area postrema syndrome	Common	Rare	Rare
Encephalopathy	Rare	Common (especially in children)	Rare
Seizures	Rare	Common with ADEM or cortical encephalitis phenotype	Rare
CSF findings			
Oligoclonal bands	<20% (transient)	<20% (transient)	>85% (persistent)
White cell count >50/μl	13% to 35%	35%	Rare
MRI of acute attacks			
Optic neuritis	Mainly posterior segments including chiasm	Long lesions (>50%), mainly anterior segments, perineural enhancement	Short lesions, mainly along the intraorbital tract
Myelitis	Longitudinally extensive (85%) single lesion; bright spotty lesions of T2 hyperintensity; elongated ring or patchy enhancement in most cases	Longitudinally extensive (60% to 80%), often coexisting with short lesions; conus frequently involved; H-sign axially; acute enhancement in 50% of patients	Multiple short lesions; periphery of cord; ring or nodular enhancement
Brain or brainstem	Often nonspecific; periventricular regions more affected, mostly area postrema; extensive white matter lesions may sometimes occur	Large "fluffy" lesions affecting both white and gray matter; cortical hyperintensity may occur; enhancement in 50% of cases and is typically nonspecific; ring enhancement is rare	Small ovoid periventricular, infratentorial, or juxtacortical T2 lesions; central vein sign is specific; ring or nodular enhancement
Initially normal brain and spine MRI	Possible (approximately 10% of brain or myelitis attacks)	Possible (up to 10% of brain or myelitis attacks)	Rare
Postattack MRI			
T2 lesion resolution	Rare	50% to 80%	Rare
New asymptomatic T2 lesion accumulation	Rare	Rare	Common*
Residual T1 hypointensity	Rare	Infrequent	Common
Persistent acute gadolinium enhancement >6 months	Rare	Rare	Rare
Clinical recovery			
Optic neuritis	Risk for poor recovery	Generally good	Generally good
Myelitis	Risk for poor recovery	Good motor recovery but risk of residual sphincter or erectile deficit	Generally good but spinal cord lesions and attacks increase the risk of disease progression
Brain or brainstem attacks	Good recovery from area postrema attacks; risk of worse recovery from attacks in other brain regions	Generally good; risk of residual cognitive deficit in patients with recurrent encephalitis	Generally good

AQP4+NMOSD v MOGAD vs MS

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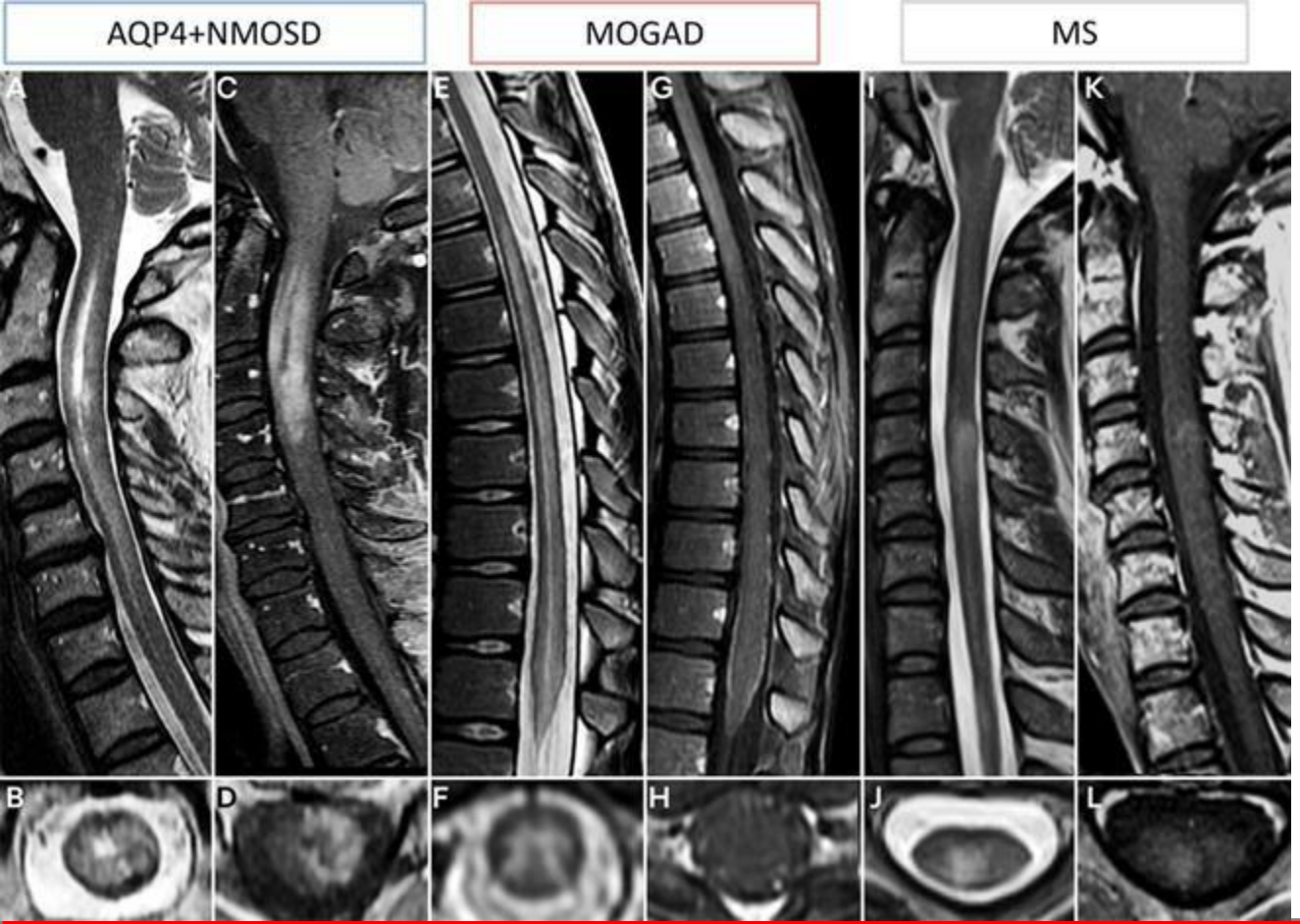


ON in AQP4+NMOSD v MOGAD v MS

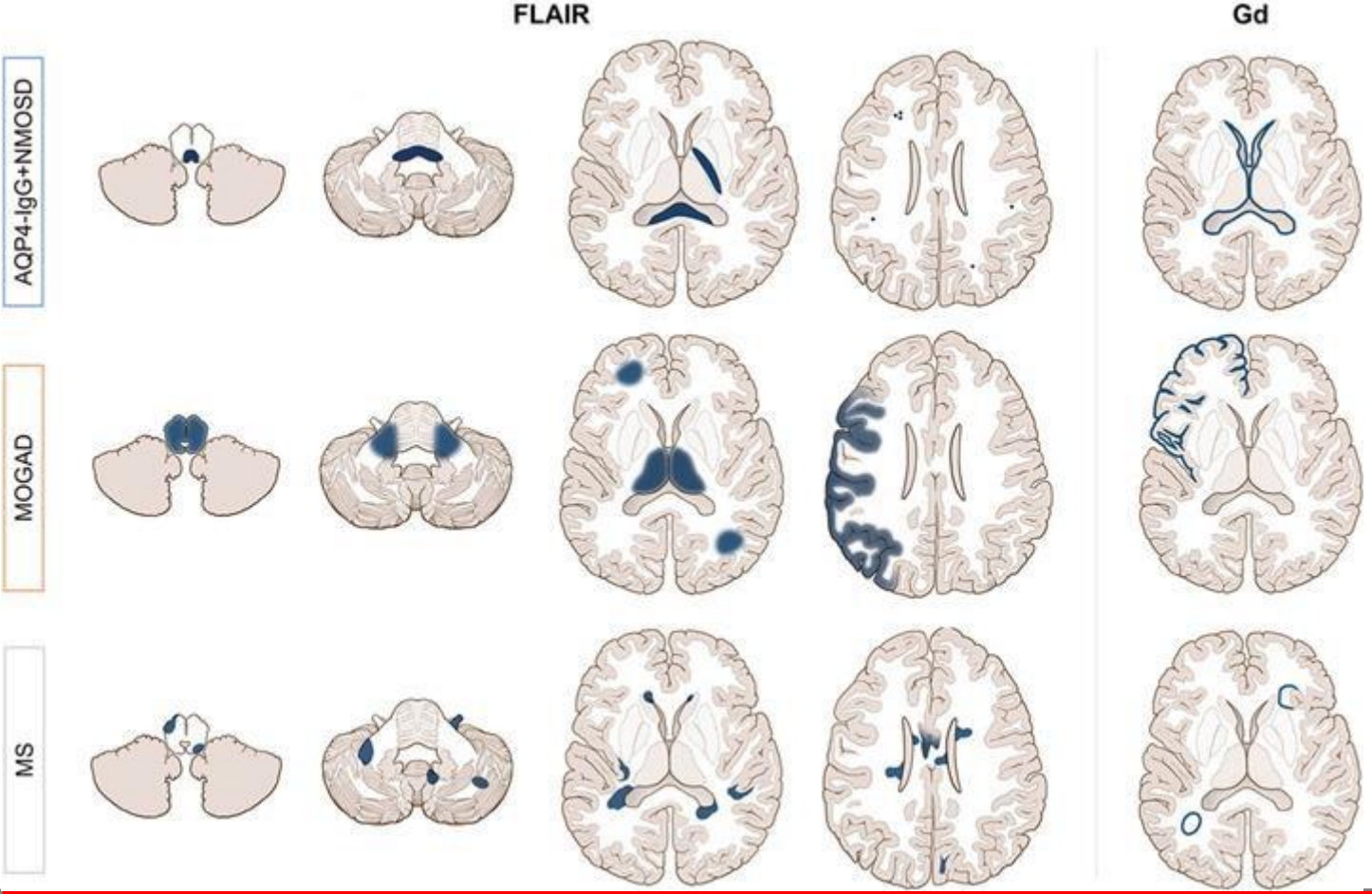


SC in AQP4+NMOSD v MOGAD v MS

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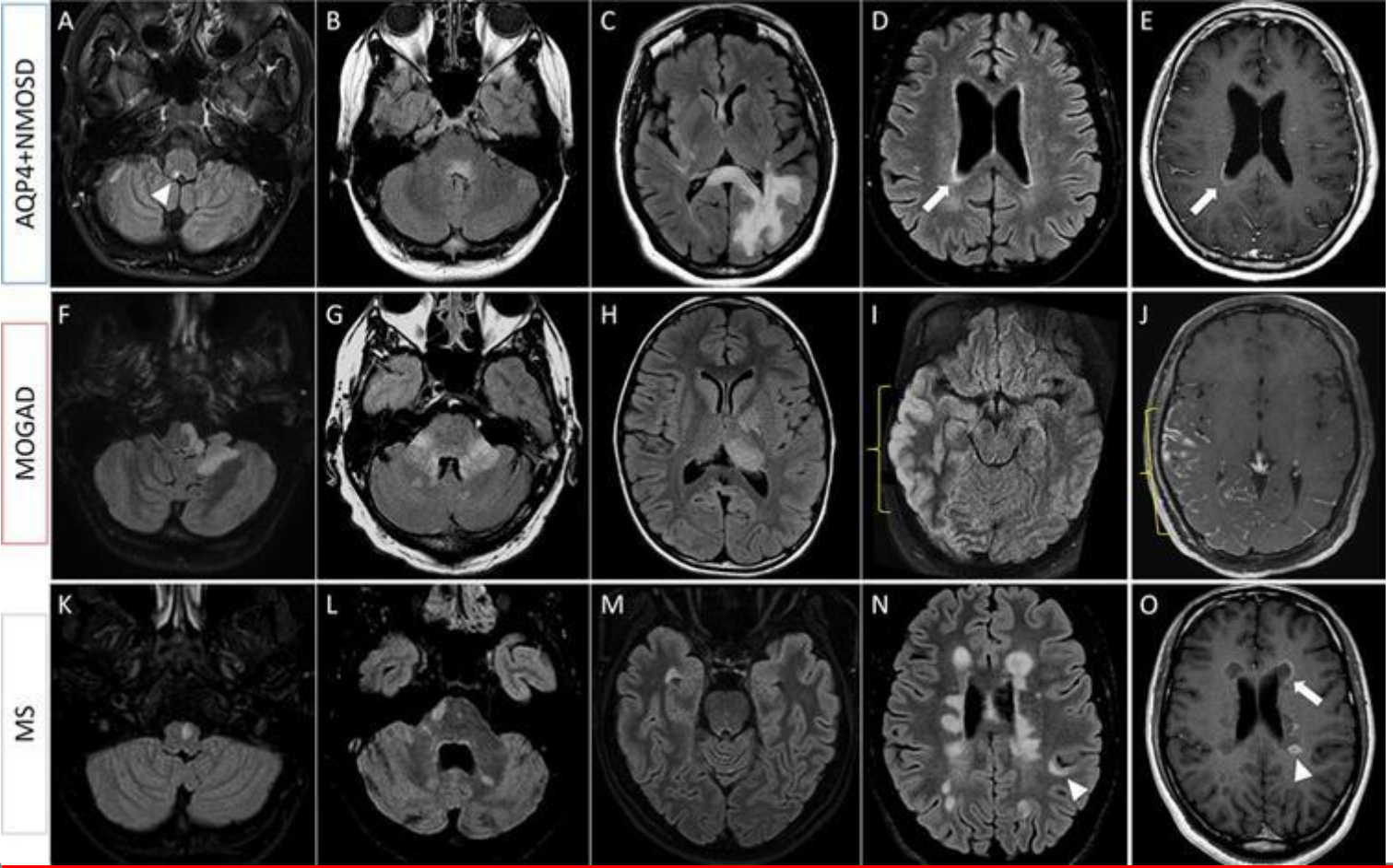


SC in AQP4+NMOSD v MOGAD v MS



Brain in AQP4+NMOSD v MOGAD v MS

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Brain in AQP4+NMOSD v MOGAD v MS

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Risk factor	Association in multiple sclerosis	Interventions in clinical practice
Vitamin D deficiency	Increased risk of multiple sclerosis (MS)/clinically isolated syndrome (CIS)[31,35,36] Increased risk of MS relapse[31,38,39]	Consider assessing serum total vitamin D level for patients with MS/CIS Definitive goal level for patients with MS not yet established; Endocrine Society guidelines recommend goal level of 30 ng/mL for the general population,[46] and some MS specialists recommend goal level of 40-60 ng/mL[47]
Smoking	Associated with increased risk of MS/CIS[58,60] and increased risk of disease progression[59,60]	Encourage cessation and educate patients regarding MS-specific smoking risks, consider referral to smoking cessation support services[64,65]
Obesity	Childhood/adolescent obesity associated with increased MS risk[66,67] Abdominal obesity associated with worse MS disability[67]	Counsel patients regarding MS-specific obesity concerns, encourage healthy weight[67]
Diet and microbiome	Gut microbiome may contribute to systemic inflammation[71,73] Unifying ideal MS diet has not yet emerged, but diet may contribute to symptom management[74-77]	Discuss potential role of higher quality diet in MS symptom management
Comorbid conditions	Systemic and psychiatric comorbidities may contribute to relapse risk,[78] degree of disability,[84,87] and adherence to treatment[85,86]	Inform patients of importance of managing comorbidities

MS Risks Factors

Number of clinical attacks	Number of lesions with objective clinical evidence	Additional data needed for a diagnosis of multiple sclerosis
≥2 clinical attacks	≥2	None[c]
≥2 clinical attacks	1 (and clear-cut historical evidence of a prior attack involving a lesion in a distinct anatomic location)[d]	None[c]
≥2 clinical attacks	1	Dissemination in space demonstrated by an additional clinical attack implicating a different central nervous system (CNS) site or by MRI
1 clinical attack	≥2	Dissemination in time demonstrated by an additional clinical attack or by MRI or CSF-specific oligoclonal bands
1 clinical attack	1	Dissemination in space demonstrated by an additional clinical attack implicating a different CNS site or by MRI AND Dissemination in time demonstrated by an additional clinical attack or by MRI OR Demonstration of CSF-specific oligoclonal bands

[b] If the 2017 McDonald criteria are fulfilled and there is no better explanation for the clinical presentation, the diagnosis is multiple sclerosis. If multiple sclerosis is suspected by virtue of a clinically isolated syndrome but the 2017 McDonald criteria are not completely met, the diagnosis is possible multiple sclerosis. If another diagnosis arises during the evaluation that better explains the clinical presentation, the diagnosis is not multiple sclerosis.

[c] No additional tests are required to demonstrate dissemination in space and time. However, unless MRI is not possible, brain MRI should be obtained in all patients in whom the diagnosis of multiple sclerosis is being considered.

[d] Clinical diagnosis based on objective clinical findings for two attacks is most secure. Reasonable historical evidence for one past attack, in the absence of documented objective neurologic findings, can include historical events with symptoms and evolution characteristic of a previous inflammatory demyelinating attack; at least one attack, however, must be supported by objective findings. In the absence of residual objective evidence, caution is needed.

Dissemination in space

- **Can be demonstrated by one or more T2-hyperintense lesions[b] that are characteristic of multiple sclerosis in two or more of four areas of the central nervous system:**
  - Periventricular
  - Cortical or juxtacortical
  - Infratentorial brain regions
  - Spinal cord

Dissemination in time

- **Can be demonstrated by:**
  - The simultaneous presence of gadolinium-enhancing and nonenhancing lesions[b] at any time
  - A new T2-hyperintense or gadolinium-enhancing lesion on follow-up MRI, with reference to a baseline scan, irrespective of the timing of the baseline MRI

2017 McDonald Criteria difference from 2010 Criteria

Cortical & juxtacortical lesions can be used for MRI DIS

Symptomatic & asymptomatic lesions can be used for DIS or DIT

CSF-specific OligoClonal Bands (OCBs) can be used for DIT with typical clinical syndrome & meeting DIS

MS Criteria

The diagnosis of AQP4-NMOSD requires the fulfillment of criteria A, B, and C

A At least one of the following core clinical manifestations of the disease

1 Acute or subacute optic neuritis

2 Acute or subacute myelitis

3 Area postrema syndrome, defined as an otherwise unexplained episode of hiccups, nausea, or vomiting lasting for ≥48 hours and that do not recede with symptomatic medications

4 Acute or subacute brainstem syndrome

5 Symptomatic narcolepsy or acute diencephalic clinical syndrome with diencephalic MRI lesions typical of AQP4-NMOSD around the third ventricle

6 Symptomatic cerebral syndrome with brain MRI lesions typical of AQP4-NMOSD (extensive white matter lesions around the lateral ventricles)

B AQP4 IgG positivity: demonstration of AQP4 IgG positivity in serum with the best available assay; cell-based assays (live cell-based assays are preferred over fixed cell-based assays) are highly recommended because of the unacceptably high risk of false-positive results with other techniques; CSF antibody testing is not routinely recommended unless serum testing is not available or might have been altered by specific treatments (eg, rituximab) and should be interpreted with caution

C Exclusion of alternative diagnoses: see TABLE 4-4 for red flags commonly associated with misdiagnosis

(AQP4+NMOSD)

The diagnosis of MOGAD requires fulfillment of A, B, and C		
A) At least one of the following core clinical manifestations of the disease	1 Acute or subacute optic neuritis	
	2 Acute or subacute myelitis	
	3 ADEM, defined as acute or subacute polyfocal neurologic deficits with encephalopathy accompanied by multifocal T2 lesions on MRI	
	4 Cerebral monofocal or polyfocal deficits without encephalopathy	
	5 Acute or subacute brainstem or cerebellar deficits	
	6 Cerebral cortical encephalitis with seizures, defined as MRI evidence of FLAIR cortical hyperintensity often with enhancement of the overlying meninges, accompanied by cerebral irritability (eg, encephalopathy, headache, focal neurologic deficit) in addition to seizures	
B) MOG IgG positivity by cell-based assay	Clear serum MOG IgG positivity, defined as: 1) at least two doubling dilutions above the assay-specific cutoff, titer cutoff, or flow cytometry ratio cutoff by live cell-based assay; or 2) titers of ≥1:100 by fixed cell-based assay	No additional supporting features required
	Low serum MOG IgG positivity, defined as low-range positivity by live cell-based assay or titers between 1:10 and 1:100 by fixed cell-based assay	The following additional items are required: 1 Negative test for AQP4 IgG 2 At least 1 supporting clinical or MRI feature
	Serum MOG IgG positivity with unknown or not reported titer	
	Isolated MOG IgG positivity in CSF	

Supporting clinical and MRI features	Optic neuritis	Bilateral simultaneous clinical involvement
		Optic nerve involvement for >50% of the nerve length on MRI
		Perineural optic sheath enhancement on MRI
		Optic disc edema
	Myelitis	Longitudinally extensive spinal cord lesion on MRI
		Central cord lesion or H-sign
		Conus lesion
	Brain or brainstem syndrome	Multiple ill-defined T2 hyperintense lesions in supratentorial and often infratentorial white matter
		Deep grey matter involvement
		Ill-defined T2 hyperintensity involving pons, middle cerebellar peduncle, or medulla
		Cortical lesion with or without lesional and overlying meningeal enhancement
C) Exclusion of alternative diagnoses including MS	See Table 4-4 for red flags commonly associated with misdiagnosis	

ADEM = Acute Disseminated Encephalomyelitis

(MOGAD)

Typical & Atypical presentations of CIS or RRMS

Typical presentations

- Acute unilateral optic neuritis
- Double vision due to an internuclear ophthalmoplegia or sixth nerve palsy[b]
- Facial sensory loss or trigeminal neuralgia[b]
- Cerebellar ataxia and nystagmus
- Partial myelopathy
- Sensory symptoms in a central nervous system pattern
- Lhermitte symptom
- Asymmetric limb weakness
- Urge incontinence or erectile dysfunction

Atypical or red flag presentations that may suggest an alternative diagnosis

- Bilateral optic neuritis or unilateral optic neuritis with a poor visual recovery
- Complete gaze palsy or fluctuating ophthalmoparesis
- Intractable nausea, vomiting, or hiccups
- Complete transverse myelopathy with bilateral motor and sensory involvement
- Encephalopathy
- Subacute cognitive decline
- Headache or meningism
- Isolated fatigue or asthenia
- Constitutional symptoms

[b] < 40 yo

Letter	Red flag MRI features
M	Meningeal enhancement
I	Indistinct lesions Increasing lesions
M	Macrobleeds Microbleeds
I	Infarcts
C	Cavities Complete ring enhancement
S	Symmetric lesions Sparing of U fibers Siderosis Spinal cord longitudinally extensive lesions (≥3 spinal cord segments)

MS Red Flags

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Atypical or red flag presentations	Laboratory and other investigations to consider
Bilateral optic neuritis or unilateral optic neuritis with poor visual recovery	Anti-aquaporin-4 (AQP4) antibody (neuromyelitis optica [NMO]-IgG), anti-myelin oligodendrocyte glycoprotein (MOG) antibody, toxic/metabolic (vitamin B_{12}, copper, etc), infectious screen (*Bartonella*, syphilis, Lyme, tuberculosis, etc), systemic autoimmune screen (erythrocyte sedimentation rate [ESR], C-reactive protein [CRP], antinuclear antibody [ANA], extractable nuclear antigen [ENA], angiotensin-converting enzyme [ACE]), genetic tests (eg, Leber hereditary optic neuropathy), optical coherence tomography
Complete gaze palsy or fluctuating ophthalmoparesis	Chest imaging (chest x-ray or CT)/positron emission tomography (PET) scan/tissue biopsy to assess for sarcoid, malignancy screen (imaging, CSF cytology, and flow cytometry), systemic autoimmune screen, infectious screen (viral, tuberculosis, Lyme, fungal, etc), additional imaging to assess for ischemic/vascular etiologies, neuromuscular etiologies
Intractable nausea, vomiting, or hiccups	Anti-AQP4 antibody (NMO-IgG), anti-MOG antibody
Complete transverse myelopathy with bilateral motor and sensory involvement	Anti-AQP4 antibody (NMO-IgG), anti-MOG antibody, systemic autoimmune screen (ESR, CRP, ANA, ENA), ACE, infectious screen (herpes viruses, mycoplasma, tuberculosis, human immunodeficiency virus [HIV], human T-cell lymphotropic virus type I [HTLV-I], etc), additional imaging (spinal magnetic resonance angiography, digital subtraction angiography) to assess for vascular causes (ischemia, vascular lesion), toxic/metabolic screen (vitamin B_{12}, copper, ceruloplasmin, etc)
Encephalopathy, subacute cognitive decline, headache or meningism, isolated fatigue or asthenia, constitutional symptoms	Infectious screen, malignancy screen (imaging, CSF cytology, and flow cytometry), systemic autoimmune/vasculitis screen (ESR, CRP, ANA, ENA, antineutrophil cytoplasmic antibodies), toxic/metabolic screen, paraneoplastic antibody panel

(MS Red Flags)

TABLE 4-4 *Red Flags* suggesting alternative diagnosis in AQP4 IgG or MOG IgG positivity

Clinical features and disease course

- Progressive disease course, either from onset (primary progressive course) or after an initial period of relapsing disease activity (secondary progressive course); consider MS, spinal cord sarcoidosis, or metabolic or genetic etiologies (eg, vitamin B_{12} deficiency, adrenoleukodystrophy)
- Hyperacute presentation (<4 hours); consider stroke or ischemic damage (eg, spinal cord infarction, ischemic optic neuropathy)
- Concomitant peripheral neuropathy; can rarely be found in patients with AQP4-NMOSD and MOGAD and its significance is unclear; consider other causes of myeloneuropathy or encephalomyeloneuropathy (eg, other autoimmune or paraneoplastic neurologic disorders, metabolic or genetic conditions)
- Comorbidities that can have similar manifestations, particularly sarcoidosis and cancer (potential for both direct and indirect CNS involvement as paraneoplastic syndrome)

CSF findings

- CSF-restricted oligoclonal bands; consider MS or sarcoidosis

MRI

- Brain abnormalities typical of MS
  - Ovoid periventricular lesions in the hemispheric white matter (perpendicularly oriented to the main ventricle axis), brainstem, or cerebellar hemispheres
  - Linear or S-shaped juxtacortical lesions
  - Ring or open-ring enhancement
- Spinal cord abnormalities typical of MS
  - Multiple, short (fewer than three contiguous vertebral body segments) lesions peripherally located on axial images (commonly along the dorsal and lateral columns), often accompanied by focal atrophy
  - Ring or open-ring enhancement
- Development of new asymptomatic brain or spine lesions over time; consider MS
- Lack of lesion resolution over time (uncommon in MOGAD)
- Central vein sign; consider MS
- Persistence of gadolinium enhancement of lesions for >6 months; consider sarcoidosis, neoplasms, or spondylotic myelopathy

Serology

- AQP4 IgG low positivity detected with ELISA
- Borderline or low MOG IgG titer; these should be interpreted based on the clinical MRI phenotype (also see supportive diagnostic criteria); consider repeat testing, preferably with a more accurate assay (eg, live versus fixed cell-based assay)

Other

- Lack of response to acute treatment with corticosteroids and plasma exchange
- Other clinical MRI findings suggestive of alternative diagnosis (eg, blood or microhemorrhages on MRI, positive screening for infections in serum or CSF)

(Red Flags)

2017 McDonald Criteria for Primary Progressive Multiple Sclerosis

One year of disability progression (retrospectively or prospectively determined) independent of clinical relapse

AND two of the following criteria:

- One or more T2-hyperintense lesions[b] characteristic of multiple sclerosis in one or more of the following brain regions: periventricular, cortical or juxtacortical, or infratentorial
- Two or more T2-hyperintense lesions in the spinal cord
- Presence of CSF-specific oligoclonal bands

[b] Unlike the 2010 McDonald criteria, no distinction between symptomatic and asymptomatic MRI lesions is required.

(PPMS)

Radiologically Isolated Syndrome (RIS) Criteria

The presence of incidentally identified central nervous system (CNS) white matter anomalies meeting the following MRI criteria:

- Ovoid, well-circumscribed, and homogeneous foci with or without involvement of the corpus callosum
- T2 hyperintensities measuring >3 mm and fulfilling Barkhof criteria[20] (at least 3 of 4) for dissemination in space
- CNS white matter anomalies

No historical accounts of remitting clinical symptoms consistent with neurologic dysfunction

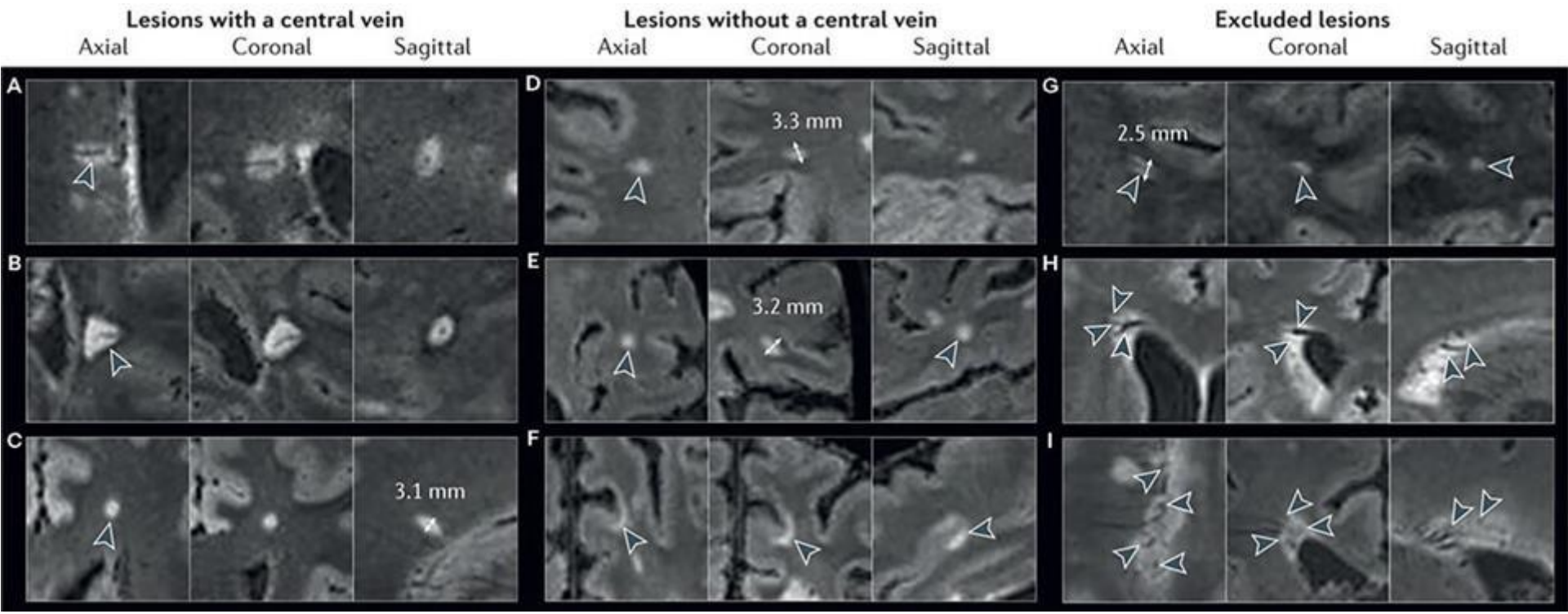
The MRI anomalies do not account for clinically apparent impairments in social, occupational, or generalized areas of functioning

The MRI anomalies are not due to the direct physiologic effects of substances (recreational drug abuse, toxic exposure) or a medical condition

Exclusion of individuals with MRI phenotypes suggestive of leukoaraiosis or extensive white matter pathology lacking involvement of the corpus callosum

The CNS MRI anomalies are not better accounted for by another disease process

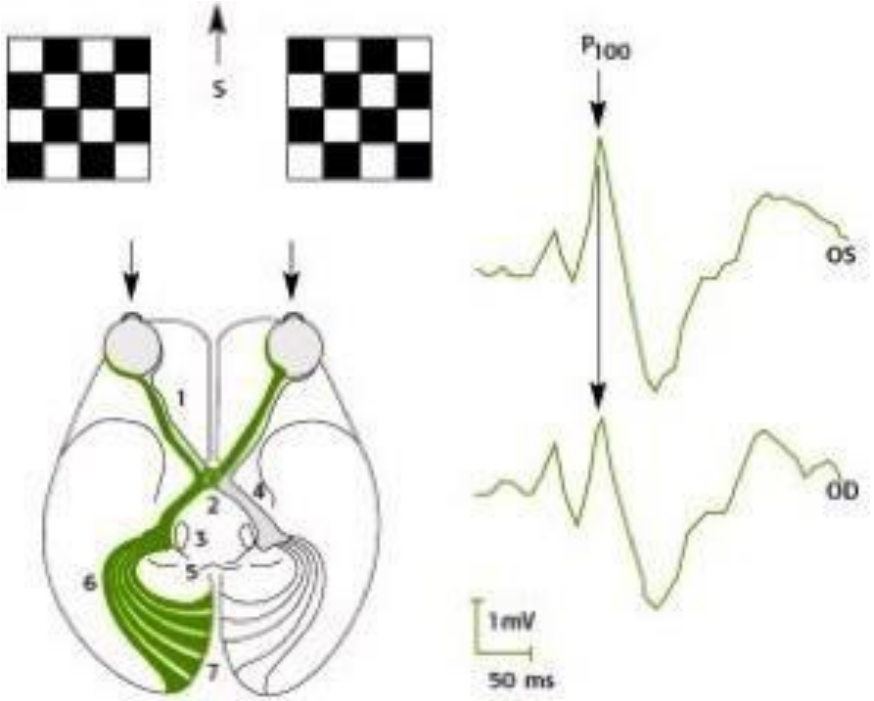
(RIS)



MS Central Vein Sign

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Visual Evoked Potentials

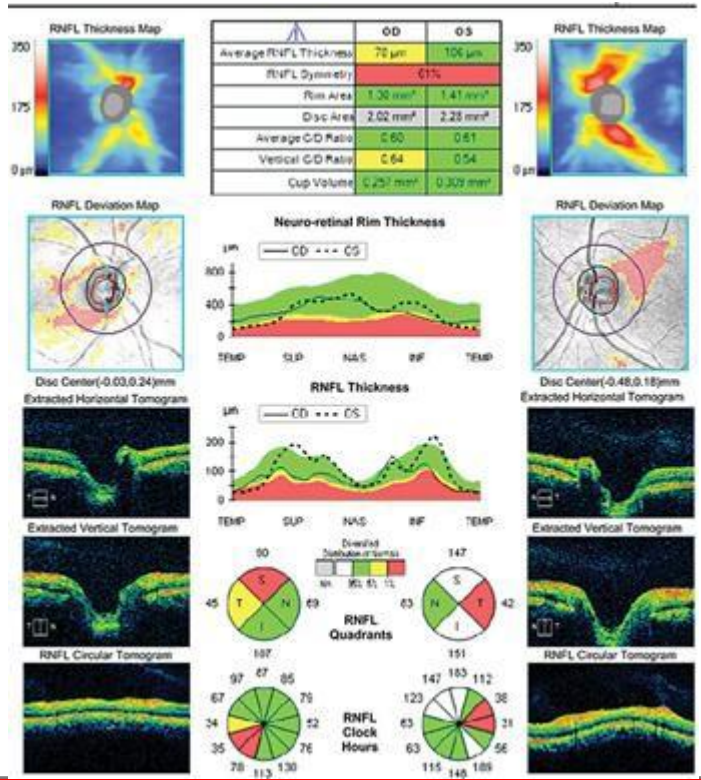


Optical Coherence Tomography = OCT

Essential Neuro-Ophthalmologist tool / retina cross-sections

Quantitative assessment axons coalescing into ON & neuronal origins

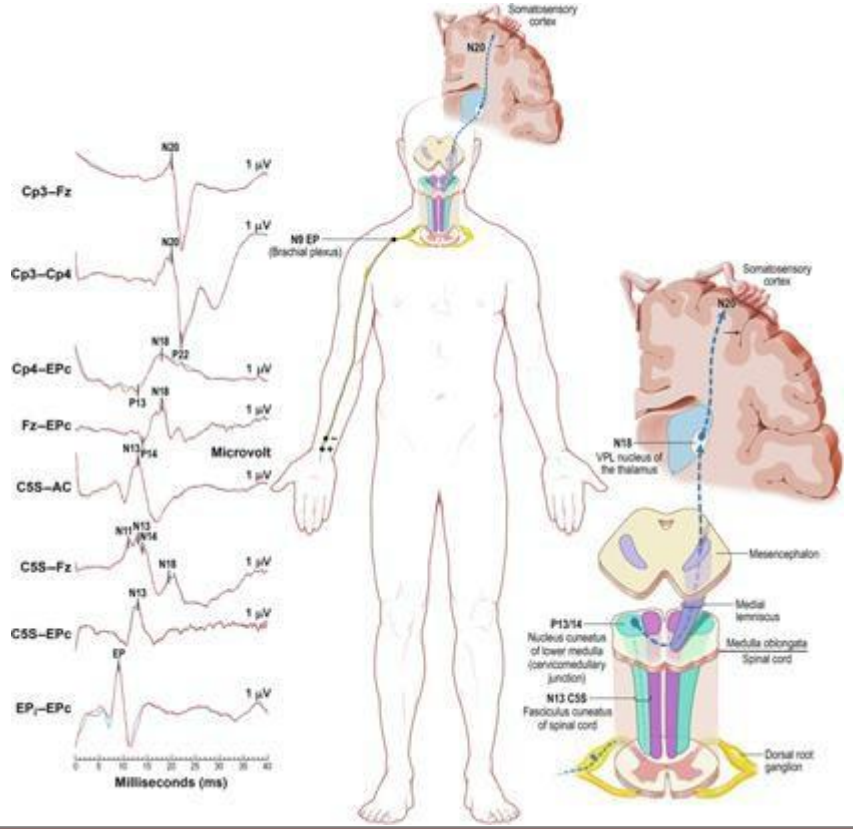
Peripapillary Retinal Nerve Fiber Layer (RNFL) thickness / Macular Ganglion Cell Layer (GCL)



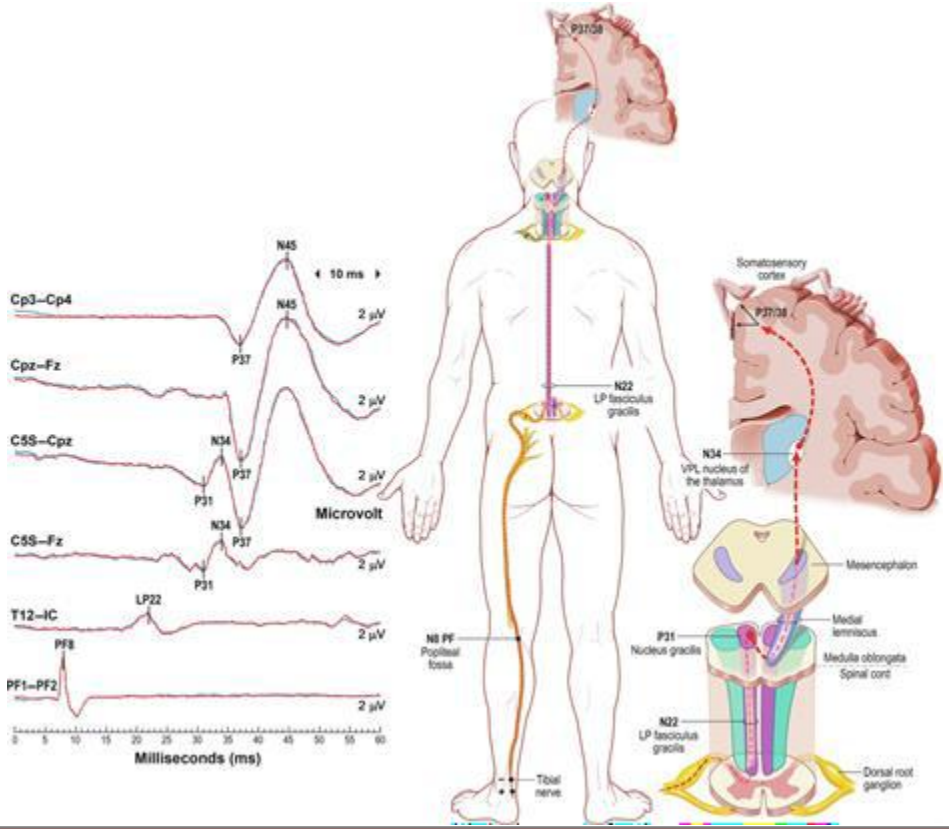
ON testing

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Median Nerve
Somatosensory Evoked Potentials (SEPs)



Posterior Tibial Nerve
Somatosensory Evoked Potentials (SEPs)



SSEPs

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Disease-modifying treatment	Subtype of multiple sclerosis	Efficacy (versus placebo, unless otherwise noted)			Mechanisms of action	Monitoring	References
		Annualized relapse rate reduction	Disability reduction	MRI			
Interferon beta-1a	Clinically isolated syndrome (CIS), relapsing-remitting multiple sclerosis (RRMS), active secondary progressive MS (SPMS)	18% for 30 µg/wk; 32% for 44 µg 3 times a week versus placebo	Interferon beta-1a 30 µg/wk and 44 µg 3 times a week reduced by 37% and 30%, respectively, versus placebo	New/persisting gadolinium-enhancing lesion or new/enlarging T2 lesion reduced 81% by 44 µg 3 times a week subcutaneously versus placebo	Inhibits proinflammatory cytokines, such as interferon gamma, tumor necrosis factor-α, and lymphotoxin; increases interleukin 10; reduced adhesion molecule and class II MHC expression	Complete blood cell count (CBC), liver function tests (LFTs), thyroid-stimulating hormone (TSH)	De Stefano N et al, 2014[26]
Interferon beta-1b	CIS, RRMS, active SPMS	31% for 250 µg every other day versus placebo	No effect in phase 3 RRMS trial; analysis of two SPMS trials found risk of 6-month confirmed disability progression reduced by 20%, and by 30-40% in active SPMS	83% reduction in rate of active lesions over 3 years versus placebo	Inhibits proinflammatory cytokines, such as interferon gamma, tumor necrosis factor-α, and lymphotoxin; increases interleukin 10; reduced adhesion molecule and class II MHC expression	CBC, LFTs, TSH	IFNB Multiple Sclerosis Study Group, 1993[27]
Glatiramer acetate	CIS, RRMS, active SPMS	29% versus placebo	28% lower disability versus placebo over 2-year trial, disability defined by 1.0 Expanded Disability Status Scale score increase	Tl-weighted gadolinium-enhancing lesions at months 6 and 12 reduced by 45% versus placebo	Alters T cell cytokine profile toward that of Th2 immunomodulatory cells; induces regulatory T cells	No laboratory monitoring	Johnson KP et al, 1995[28]
Teriflunomide	CIS, RRMS, active SPMS	31-36% reduction versus placebo at 14 mg/d	30% and 31% reduction in 12-week confirmed disability progression versus placebo	80% reduction in gadolinium-enhancing lesions versus placebo at 14-mg dose	Inhibits dihydroorotate dehydrogenase, thereby inhibiting lymphocyte proliferation	LFTs, blood pressure clinical monitoring for peripheral neuropathy	Confavreux C et al, 2014[29]
Fumarates							
Dimethyl fumarate	CIS, RRMS, active SPMS	44-53% reduction versus placebo	Confirmed disability progression proportion reduced by 38% (significant) and 21% (nonsignificant)	Gadolinium-enhancing lesions reduced by 90% and 74%	Activates nuclear factor erythroid 2-related factor 2 (Nrf2) pathway, which enhances response to oxidative stress	CBC with differential, LFTs	Gold R et al, 2012[30]; Fox RJ et al, 2012[31]
Diroximel fumarate and monomethyl fumarate	CIS, RRMS, active SPMS	Diroximel fumarate and monomethyl fumarate have the same active metabolite as dimethyl fumarate; diroximel fumarate and monomethyl fumarate were approved at doses yielding an equivalent amount of the active metabolite to dimethyl fumarate	Not studied	Not studied	Activates nuclear factor erythroid 2-related factor 2 (Nrf2) pathway, which enhances response to oxidative stress	CBC with differential, LFTs	Package inserts[32,33]

(MS DMTs)

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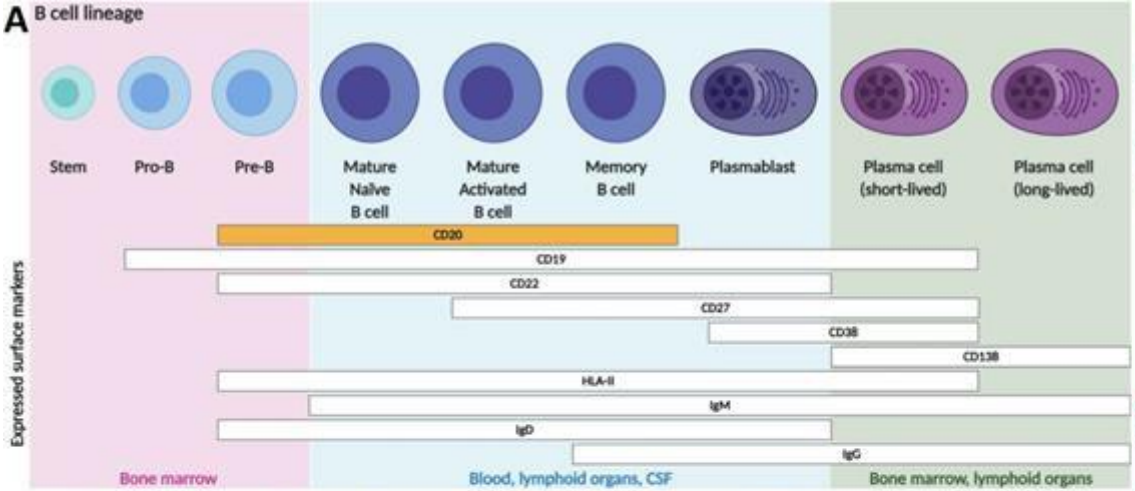
Disease-modifying treatment	Subtype of multiple sclerosis	Efficacy (versus placebo, unless otherwise noted)			Mechanisms of action	Monitoring	References
		Annualized relapse rate reduction	Disability reduction	MRI			
Sphingosine-1-phosphate receptor modulators							
Fingolimod	CIS, RRMS, active SPMS in adults and children aged 10 years and older	50% versus placebo;52% versus interferon beta-1a 30 µg/ week 82% versus interferon beta-1a 30 µg/wk in children 10-17 years old	30% confirmed disability progression reduction versus placebo in adults; no difference versus interferon beta-1a 30 µg/wk in children older than 12 months	30% reduction in rate of brain volume reduction versus placebo or interferon beta-1a 30 µg/wk in adults and 66% reduction versus interferon beta-1a 30 µg/wk in gadolinium-enhancing lesions in children 10-17 years old	Modulates sphingosine-1-phosphate receptors 1, 3, 4, and 5; must be phosphorylated before being active in vivo; lymphocytes unable to migrate out of lymphoid tissue; may have direct effects in central nervous system (CNS)	CBC with differential, LFTs, ophthalmology, dermatology, blood pressure	Cohen JA et al, 2010[34]; Kappos L et al, 2010[35]; Chitnis T et al, 2018[36]
Siponimod	CIS, RRMS, active SPMS	55% versus placebo in SPMS	21% 3-month confirmed disability progression reduction versus placebo in SPMS	79% reduction in T2 lesion volume	Modulates sphingosine-1-phosphate receptors 1 and 5; lymphocytes are unable to migrate out of lymphoid tissue; may have direct effects in CNS	CBC with differential, LFTs, ophthalmology, dermatology, blood pressure	Kappos L et al, 2018[37]
Ozanimod	CIS, RRMS, active SPMS	38% and 48% reduction versus interferon beta-1a 30 µg/wk	No statistical difference for confirmed disability progression versus interferon beta-1a 30 µg/wk	53% and 63% reduction in gadolinium-enhancing lesions versus interferon beta-1a 30 µg/wk	Modulates sphingosine-1-phosphate receptors 1 and 5; lymphocytes are unable to migrate out of lymphoid tissue; may have direct effects in CNS	CBC with differential, LFTs, ophthalmology, dermatology, blood pressure	Cohen JA et al, 2019[38]; Comi G et al, 2019[39]
Ponesimod	CIS, RRMS, active SPMS	30.5% versus teriflunomide 14 mg/d	No statistical difference for confirmed disability progression versus teriflunomide	New T1-weighted gadolinium-enhancing lesions reduced by 58.5% versus teriflunomide	Modulates sphingosine-1-phosphate receptor 1; lymphocytes are unable to migrate out of lymphoid tissue; may have direct effects in CNS	CBC with differential, LFTs, ophthalmology, dermatology, blood pressure	Kappos L et al, 2021[40]
Oral cladribine	RRMS and active SPMS, after suboptimal response to one or more disease-modifying therapy	58% versus placebo	Nonsignificant reduction in 3-month confirmed disability progression	Median T1-weighted gadolinium-enhancing lesions and active T2-weighted lesions reduced	Cytotoxic to T and B lymphocytes by impairing DNA synthesis	CBC with differential, LFTs	Giovanni G et al, 2010[41]

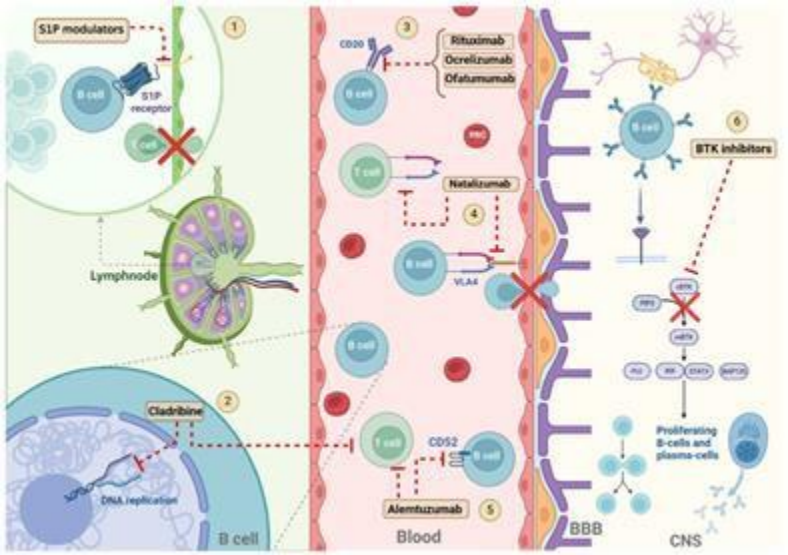
(MS DMTs)

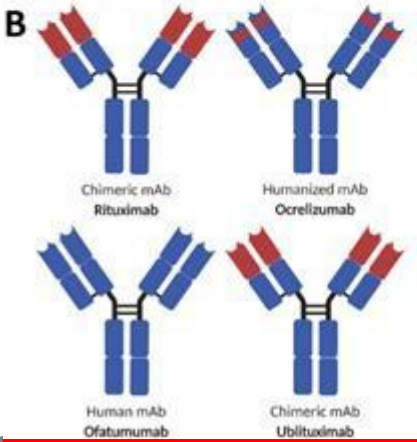
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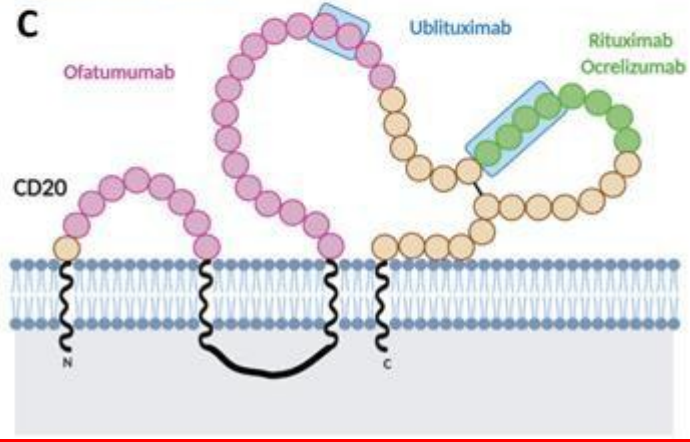
Disease-modifying treatment	Subtype of multiple sclerosis	Efficacy (versus placebo, unless otherwise noted)			Mechanisms of action	Monitoring	References
		Annualized relapse rate reduction	Disability reduction	MRI			
Natalizumab	CIS, RRMS, and active SPMS	67% versus placebo	42% reduction in 12-week confirmed disability progression	92% reduction in T1-weighted gadolinium-enhancing lesions	Monoclonal antibody targeting the alpha 4 integrins, part of the VLA-4 adhesion molecule	LFTs and JC virus antibody	Polman CH et al, 2006[42]
Anti–CD20-ocrelizumab	CIS, RRMS, active SPMS, primary progressive MS	In relapsing MS: 46% and 47% compared with interferon beta-1a 44 µg 3 times a week subcutaneously	In relapsing MS: 40% reduction in 12-week confirmed disability progression versus interferon beta-1a; in primary progressive MS: 24% reduction in 12-week confirmed disability progression versus placebo	94% and 95% reduction in T1-weighted gadolinium-enhancing lesions versus interferon beta-1a 44 µg 3 times per week	Lytic monoclonal antibody targeting CD20 molecule on surface of B cells	CBC with differential, lymphocyte subsets, quantitative immunoglobulins, LFTs	Hauser SL et al, 2017[43]; Montalban X et al, 2017[44]
Anti–CD20-ofatumumab	CIS, RRMS, active SPMS	51% and 59% compared with teriflunomide 14 mg/d	34.4% reduction in 12-week confirmed disability progression versus teriflunomide	98% and 94% reduction in T1-weighted gadolinium-enhancing lesions versus teriflunomide 14 mg daily	Lytic monoclonal antibody targeting CD20 molecule on surface of B cells	CBC with differential, lymphocyte subsets, quantitative immunoglobulins, LFTs	Hauser SL et al, 2020[45]
Anti–CD52-alemtuzumab	RRMS and active SPMS, after suboptimal response to two or more other agents	49-55% reduction versus interferon beta-1a 44 µg 3 times a week	42% reduction in confirmed disability progression in study 1, no significant reduction in study 2 versus interferon beta-1a 44 µg 3 times per week	Nonsignificant reduction in T2-weighted lesion volume versus interferon beta-1a 44 µg 3 times per week	Monoclonal antibody that lyses cells expressing CD52	CBC with differential, lymphocyte subsets, LFTs, TSH, creatinine, urinalysis	Cohen JA et al, 2012[46]; Coles AJ et al, 2012[47]
Mitoxantrone	SPMS, worsening RRMS; note: 3 randomized controlled trials had only 287 subjects with MS in total	66% reduction versus placebo in RRMS	64% reduction in 3-month confirmed disability progression in worsening RRMS and SPMS	59% reduction in new lesions over 2 years in RRMS	Anthracenedione chemotherapeutic agent causing myelosuppression	CBC with differential, LFTs, cardiac ejection fraction	Millefiorini E et al, 1997[48]; Hartung HP, et al, 2002[49]

(MS DMTs)



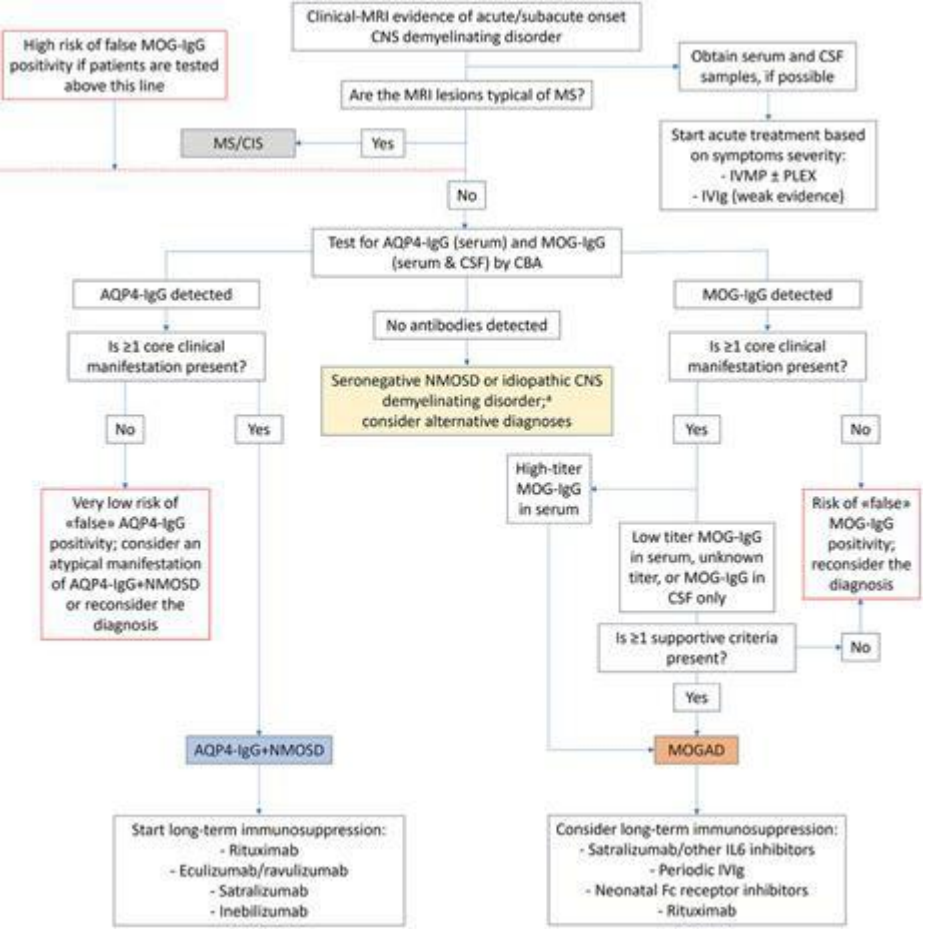






Humanized **monoclonal antibodies** reduce relapse risk, disability progression, & new /active MRI lesions; humanized (ocrelizumab, ofatumumab), mouse-human chimeric (ublituximab, rituximab). **Anti-CD20** Abs bind to **B cell** surface CD20 protein => cell death via **Antibody-Dependent Cellular Toxicity (ADCC)** by other immune cells like natural killer cells or **Complement-Dependent Cytotoxicity (CDC)** complement system activation => membrane attack complex => holes in B cell membrane

MS DMT



Rituximab IV CD20 monoclonal antibody commonly used for AQP4-NMOSD, effective relapse prevention in ~60-70% cases, less effective in MOGAD

Eculizumab / Ravulizumab IV monoclonal antibodies directed against C5 complement protein inhibiting final steps of complement pathway, shown to be highly effective for relapse prevention in AQP4-NMOSD, not studied in MOGAD

Satralizumab SQ IL-6 receptor monoclonal antibody proven effective for AQP4-NMOSD (being studied for relapse prevention in MOGAD)

Tocilizumab IV IL-6 receptor inhibitor proven superior to azathioprine in preventing AQP4-NMOSD relapses

(NMOSD & MOGAD Treatment)

Aquaporin-4 (AQP4) IgG considered directly pathogenic & emerging studies suggest **Myelin Oligodendrocyte Glycoprotein (MOG) IgG** potentially is pathogenic - a major difference from **MS** which lacks a specific antibody biomarker & has substantially different pathophysiologic mechanisms leading to CNS damage

AQP4 IgG & MOG IgG predominantly **IgG1 subclass** - significantly more efficient **complement activation** by AQP4 IgG than MOG IgG

AQP4+NMOSD 9:1 **females**: males (9:1 ratio) & different prevalence across racial & ethnic groups (diagnostic access or social determinants of health)

AQP4 IgG short myelitis lesions 14% of cases

AQP4+NMOSD most common brain manifestation -**area postrema syndrome** intractable nausea, vomiting or hiccups

MOGAD associated **ON (Optic Neuritis)** most with intense orbital or frontal pain in days before typical bilateral onset, can lead to blindness over hours to a few days

MOGAD MRI initially normal up to 10% despite severe brain &/or spinal cord dysfunction

MOGAD **cerebral cortical encephalitis** - subacute seizures & other irritation (eg, headache, encephalopathy, focal neurologic deficits), cortical FLAIR hyperintensity MRI

AQP4+NMOSD typically relapsing course after presenting attack, **~ ½ MOGAD monophasic** without relapses.

MOGAD MRI lesions resolve completely ~50-70% of cases / AQP4 IgG more destructive attacks typically residual T2 hyperintensity & atrophy of surrounding parenchyma.

Natural history of **MS** primary or secondary progressive disease course is extremely rare in AQP4-NMOSD & MOGAD

AQP4-NMOSD & MOGAD accurate diagnosis requires AQP4 IgG or MOG IgG positivity with reliable assay + compatible clinical MRI phen otype

CSF testing should be considered if seronegative with strong diagnostic suspicion of MOGAD, because isolated CSF MOG IgG posi tivity may rarely occur

AQP4 IgG ~100% specificity for both fixed & live cell-based assays, minimal false-positive risk

Cell-based high-titer MOG IgG positivity highly predictive of MOGAD, false positivity is typically low titer serum & SCF, low pretest probability screening not recommended

Prompt acute-treatment initiation for disease attacks is fundamental in AQP4-NMOSD & MOGAD - delay associated with poorer recovery & worse outcomes.

To avoid symptom recrudescence, consider **oral corticosteroid** taper after acute treatment of MOGAD attacks

Consider early **plasma exchange** with CNS demyelination acute attack presenting with severe disability, regardless of antibody serostatus

MOG IgG titer periodic measurements may be informative - seronegative conversion reduces future relapse risk

Rituximab is commonly used worldwide for AQP4-NMOSD - relapse prevention ~60-70% cases

IVIg reasonable option for pediatric MOGAD attack prevention when preferable to avoid immunosuppression given higher risk of infections

(NMOSD & MOGAD Key Points)

Epstein-Barr Virus (EBV) causative role, necessary but insufficient trigger; mechanism unclear (? molecular mimicry…)

Earliest stage high CSF T-cell specific EBV infected cells; higher blood levels active EBV genes in active symptoms; antiviral therapy unproven

Proposed McDonald Criteria Updates:

ON 5th site - increases sensitivity, reduces specificity

DIS ⅗: ON, periventricular, juxtacortical/cortical, SC, infratentorial

DIT: GAD enhancing & non-enhancing lesions in single MRI or lesion accumulation by serial MRIs

If no DIT by MRI - MS if ⅓ +CSF only (don't need paired serum) kFLCs (kappa free light chains) = OCB (Oligo-Clonal Bands)

Central Vein Sign = MRI T2 thin hypodense line/dot in center of white matter plaque (6 lesions or 50% i f < 6)

At least ⅘ sites involved

Paramagnetic Rim Lesions (PRLs): some chronic lesions have persistent demyelination - microglia & macrophages accumulate Fe

hi spec 99.7%, low sensitivity 24%, ≥ 4 earlier motor & cognitive disability

Disease-Modifying Therapy (DMT) selecting optimal treatment for newly diagnosed challenging,

Shifting to early hi-efficacy agents - RCTs relative disability reduction vs moderate efficacy agents

CD20 B cell therapies FDA approved

q 6 months **Ocrelizumab** (Ocrevus) IV/SQ or **Ublituximab** (Briumvi) or monthly **Ofatumumab** (Kesimpta) vs

not approved ? cheaper **Rituximab** q6m

Ocrelizumab safe immediately prior to pregnancy & breastfeeding

Experimental therapies for PIRA (Progression Independent of Relapse Activity) - associated with neurodegeneration & worse outcomes

Low **vitamin D** increased risk or MS & activity; RCTs no difference in relapse with hi vs low 25OHD (25-hydroxy vitamin D)

Radiologically Isolated Syndrome (RIS) - incidental on MRI - majority (about 80%) will have clinical event w/in 15 years

DIS + DIT / CVS / CSF kFLCs or OCBs

From Kristin Galetta, MD, Assistant Professor Neurology, Stanford University, AAN Fall Conference

ON inaugural event 25-35% MS, subsequent manifestation 70%, still need ≥ 2 other areas involved

MRI, OCT, VEP - improves sensitivity / reduces specificity

MS mimics - misdiagnosis rate ~ 20%

CVS (MRI T2, SWI) MS plaques typically perivenular, ? more important children & > 50 yo

Intrathecal synthesis IgG central concept in MS - OCBs gold standard - substitute for DIT; light chains made in 10-40% excess of heavy chains

k-FLC (kappa-free light chains) comparable Sn & Sp to OCBs; IgG but also IgA, IgM, easier/faster, cheaper, quantitative, rater-independent

DIT excludes monophasic disease (especially ADEM)

RIS - 50% experience 1st clinical event w/in 10 years, increased probability < 37, M, infratentorial or SC, Gad+, CSF unique OCBs

DMF (Dimethyl fumerate, Tecfidera) 80% risk reduction 1st event over 2 years

Teriflunomide (Aubagio) - 62% reduction demyelinating events

Autoimmune encephalitis (AE)

Subacute (< 3 months) AMS / working memory / Psych; at least 1 new focal CNS, new seizures, CSF pleocytosis, MRI, reasonable exclusion other (NDD, FND, Psych)

Autoimmune Limbic Encephalitis (ALE) vs Viral Encephalitis (VE)

Favor VE > 45, acute, fever, headache, N/V, CSF pleocytosis

Favor ALE < 45, subacute, Psych/memory, SE

Neurosarcoidosis clinicoradiographic phenotypes:

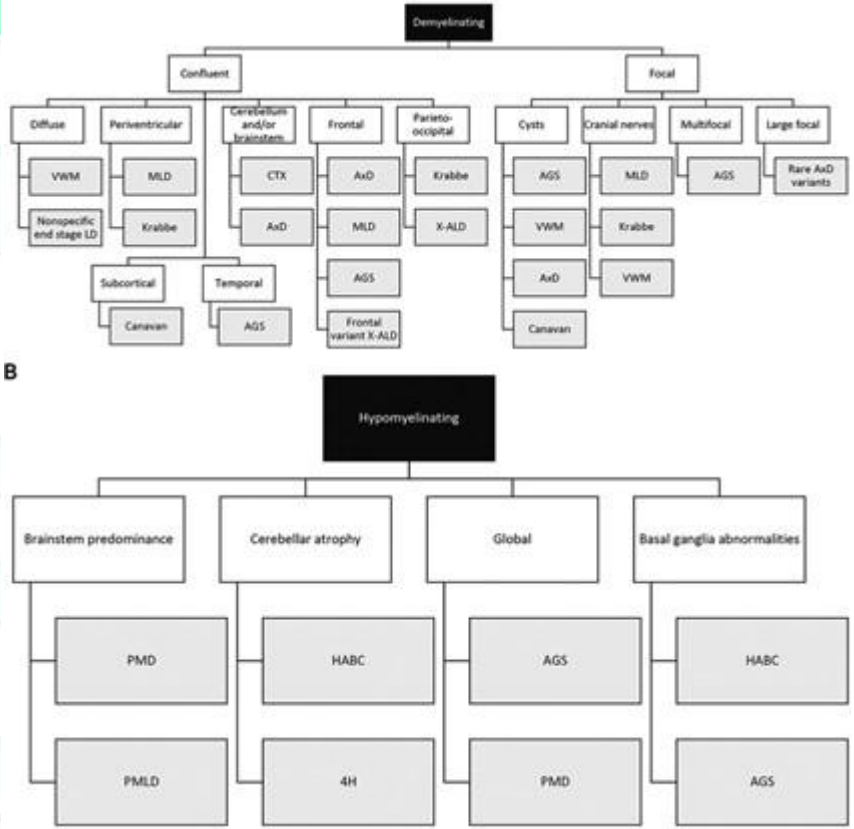
Hydrocephalus (7-14%), meingoventricular inflammation (4th, basal cisterns), may need CSF diversion

Trigeminal involvement (5-12%)

Non-enhancing T2/FLAIR white matter lesions - bifrontal, subcortical, lower lesion load & less CC involvement c/w MS

Spencer Hutto, MD, Assistant Professor Neurology, Emory University School of Medicine, AAN Fall Conference 2024

Leukodystrophy	Gene	Forms/subtypes	Diagnostic pearls
Adrenoleukodystrophy	*ABCD1*	Cerebral adrenoleukodystrophy, adrenomyeloneuropathy, Addison disease (adrenal insufficiency)	Cerebral adrenoleukodystrophy can present in childhood with a change in fine motor skills, with new-onset behavior and attention issues
Metachromatic leukodystrophy	*ARSA*	Late infantile, early juvenile, late juvenile, adult	Typically presents in toddler years with difficulty walking but can present as a peripheral neuropathy or with involvement of the cranial nerves
Aicardi-Goutières syndrome	*TREX1, RNASEH2A, RNASEH2B, RNASEH2C, SAMHD1, ADAR1, IFIH1, LSM11, RNU7-1*	Neonatal, late onset, isolated spastic paraparesis	Mimic of congenital infection; can present in early toddler years with subacute onset of new gross motor and tone abnormalities; neurologic disability is often accompanied by signs and symptoms of systemic inflammation
Alexander disease	*GFAP*	Neonatal, infantile, juvenile, adult; type I, type II	Can be associated with early seizures and feeding difficulties
Krabbe disease	*GALC*	Early infantile, late infantile, late onset	Can present with extreme irritability with spasticity
Pelizaeus-Merzbacher disease	*PLP1*	Connatal, classic, null, spastic paraplegia 2	Can present with stridor and nystagmus
Vanishing white matter disease	*EIF2B1, EIF2B2, EIF2B3, EIF2B4, EIF2B5*	Antenatal, infantile (includes Cree leukoencephalopathy), early childhood, late childhood/juvenile-adult	Can present after a head injury or stressor with pronounced altered mental status with vomiting and hypotonia
***POLR3*-related leukodystrophy**	*POLR3A, POLR3B, POLR1C*	Hypomyelination, hypodontia, hypogonadotropic hypogonadism (4H syndrome)	Neurologic findings often accompanied by dental and endocrine abnormalities
***TUBB4A*-related disorders**	*TUBB4A*	Hypomyelination with atrophy of basal ganglia and cerebellum, isolated hypomyelination, whispering dysphonia	Can present with delayed development followed by loss of walking ability in later childhood
Cerebrotendinous xanthomatosis	*CYP27A1*		Systemic complications can precede neurologic findings
Canavan disease	*ASPA*	Neonatal/infantile, mild/juvenile	Can be associated with extreme hypotonia, macrocephaly, and exaggerated startle response



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Adrenoleukodystrophy (ALD) 3 distinct phenotypes

Cerebral Adrenoleukodystrophy

typically presents with attention & behavioral changes in school-age boys

Adrenomyeloneuropathy (AMN)

Adrenal Insufficiency (Addison only)

Metachromatic Leukodystrophy (MLD) - most commonly decline in gross motor function in early childhood; peripheral neuropathy or strabismus can precede

Aicardi-Goutières Syndrome - multiorgan autoinflammatory disorder with neurologic disability

Early-onset mimics congenital infection

Later-onset can present like **Acute Disseminated EncephaloMyelitis (ADEM)**

Type I Alexander Disease - macrocephaly, developmental delay, feeding issues, vomiting

Type II Alexander Disease - autonomic dysfunction, eye movement abnormalities, prominent bulbar symptoms

Krabbe Disease - blood psychosine diagnostic biomarker

Pelizaeus-Merzbacher Disease - may present with prominent nystagmus or stridor

Vanishing White Matter Disease - imaging findings often precede neurologic stochastic decline which can be triggered by head trauma or infection

***POLR3*-Related Leukodystrophy** - neurologic features often accompanied by endocrine abnormalities & abnormal dentition

***TUBB4A*-Related Disorders** - in most cases cognition less affected than motor function

Cerebrotendinous Xanthomatosis - systemic signs & symptoms often precede neurologic dysfunction

Canavan Disease & Alexander Disease - both can have macrocephaly



Shifting Gears

Features Consistent With Any Central Nervous System Localization

- Brisk reflexes in an affected limb
- Increased tone in an affected limb
- Sensory or motor symptoms limited to two limbs on the same side of the body
- Weakness of extensors in the upper extremity
- Weakness of flexors in the lower extremity

Features Consistent With a Spinal Cord Localization

- Sensory loss in both legs with a sensory level on the trunk
- Decreased pinprick sensation on one side, weakness and diminished proprioception on the other side

Features Consistent With a Peripheral Localization

- Axial or limb pain
- Diminished reflexes in an affected limb
- Weakness and diminished pain sensation in the same limb are not due to a spinal cord lesion and are due to either a peripheral or brain/brainstem lesion
- Sensory loss that involves the hands and feet but not the trunk

Structure	Principal Muscles Involved	Sensory Distribution	Diminished Reflexes
C5 root	Rhomboids[b] Supraspinatus[b] Infraspinatus[b] Deltoid[b] Brachioradialis[b] Biceps	Lateral upper arm	Biceps, brachioradialis
C6 root	Supraspinatus Infraspinatus Deltoid Brachioradialis[b] Triceps Flexor carpi radialis[b] Extensor carpi radialis[b] Pronator teres[b]	Lateral forearm and hand, first two digits	Biceps, brachioradialis
C7 root	Triceps[b] Flexor carpi radialis[b] Extensor carpi radialis[b] Pronator teres[b] Extensor digitorum[b] Flexor digitorum[b] Flexor pollicis longus[b]	Centermost portion of the distal dorsal forearm and third digit	Triceps
C8 root	Triceps Extensor indicis[b] Flexor pollicis longus[b] Abductor pollicis brevis[b] Interossei[b]	Fourth and fifth digits and medial hand	None
T1 root	Abductor pollicis brevis[b] Interossei[b]	Medial forearm	None

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[b] Muscle may be more prominently involved.

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Structure	Principal Muscles Involved	Sensory Distribution	Diminished Reflexes
L2 root	Iliopsoas[b] Adductor longus[b]	Anterior thigh	None
L3 root	Iliopsoas[b] Quadriceps[b] Adductor longus[b]	Medial thigh, knee	Patellar
L4 root	Quadriceps[b] Adductor longus[b] Tibialis anterior	Medial lower leg	Patellar
L5 root	Gluteus medius[b] Hamstrings Tibialis anterior[b] Extensor digitorum[b] Peroneus longus[b] Tibialis posterior[b]	Lateral lower leg, dorsal foot	None
S1 root	Gluteus maximus[b] Hamstrings[b] Gastrocnemius[b] Peroneus longus Extensor digitorum Tibialis posterior	Posterior/lateral lower leg, plantar aspect of foot	Ankle

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[b] Muscle may be more prominently involved.

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Structure	Principal Muscles Involved	Sensory Distribution	Diminished Reflexes
Upper trunk	Supraspinatus[b] Infraspinatus[b] Deltoid[b] Brachioradialis[b] Biceps[b] Triceps Pronator teres Extensor carpi radialis Flexor carpi radialis	Lateral upper arm and forearm and first one to two digits	Biceps, brachioradialis
Lower trunk	Triceps Extensor digitorum[b] Extensor indicis[b] Flexor digitorum profundus[b] Flexor pollicis longus[b] Abductor pollicis brevis[b] Interossei[b]	Medial forearm and hand, fourth and fifth digits	None
Middle trunk	Triceps[b] Flexor carpi radialis[b] Extensor carpi radialis[b] Pronator teres[b] Extensor digitorum[b] Flexor digitorum[b] Flexor pollicis longus[b]	Centermost portion of the distal forearm and third digit	Triceps
Lateral cord	Biceps[b] Pronator teres[b] Flexor carpi radialis[b] Flexor digitorum[b] Flexor pollicis longus[b]	Lateral forearm, palmar hand and first three digits	Biceps
Medial cord	Flexor digitorum[b] Flexor pollicis longus[b] Abductor pollicis brevis[b] Interossei[b]	Medial forearm and hand, fourth and fifth digits	None
Posterior cord	Deltoid[b] Teres minor[b] Brachioradialis[b] Triceps[b] Extensor carpi radialis[b] Extensor digitorum[b] Extensor indicis[b]	Dorsal aspect of upper arm, forearm, lateral hand, and the first three digits	Triceps, brachioradialis

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[b] Muscle may be more prominently involved.

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Structure	Principal Muscles Involved	Sensory Distribution	Diminished Reflexes
Lumbar plexus	Iliopsoas Quadriceps Adductor longus	Anterior thigh, medial and lateral thigh, medial lower leg	Patellar
Lower lumbosacral plexus	Gluteus medius Gluteus maximus Hamstrings Tibialis anterior Extensor digitorum Peroneus longus Tibialis posterior Gastrocnemius	Posterior thigh, lateral lower leg, all of foot	Ankle

Structure	Principal Muscles Involved	Sensory Distribution	Diminished Reflexes
Dorsal scapular	Rhomboids[b]	None	None
Suprascapular	Supraspinatus[b] Infraspinatus[b]	None	None
Musculocutaneous	Biceps[b]	Lateral forearm	Biceps
Axillary	Deltoid	Lateral shoulder	None
Radial			
Superficial sensory nerve at the wrist	None	Dorsal hand and first three digits	None
Posterior interosseous	Extensor digitorum[b] Extensor indicis[b]	None	None
At the spiral groove	Brachioradialis[b] Extensor carpi radialis[b] Extensor digitorum[b] Extensor indicis[b]	Dorsal hand and first three digits	Brachioradialis
At the axilla	Triceps[b] Brachioradialis[b] Extensor carpi radialis[b] Extensor digitorum[b] Extensor indicis[b]	Dorsal aspect of upper arm, forearm, lateral hand, and the first three digits	Brachioradialis, triceps
Median			
Anterior interosseous	Flexor pollicis longus[b] Flexor digitorum profundus 1-2[b]	None	None
At the carpal tunnel	Abductor pollicis brevis[b]	Palmar first through third digits	None
At the elbow	Pronator teres[b] Flexor carpi radialis[b] Flexor pollicis longus[b] Flexor digitorum profundus 1-2[b] Abductor pollicis brevis[b]	Palmar hand and first through third digits	None
Ulnar			
At the wrist	Interossei[b] Interossei[b]	Palmar aspect of fourth and fifth digits	None
At the elbow	Flexor digitorum profundus 4-5[b]	Medial hand, fourth and fifth digits	None

[a] Modified with permission from London ZN.[1] © 2019 American Academy of Neurology.
[b] Muscle may be more prominently involved.

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Structure	Principal Muscles Involved	Sensory Distribution	Diminished Reflexes
Femoral			
Above inguinal ligament	Iliopsoas Quadriceps	Medial thigh and lower leg	Patellar
Below inguinal ligament	Quadriceps	Medial thigh and lower leg	Patellar
Obturator	Adductor longus	Medial thigh	None
Sciatic	Hamstrings Tibialis anterior Extensor digitorum Peroneus longus Tibialis posterior Gastrocnemius	Lateral lower leg, all of foot	Ankle
Fibular (peroneal)			
Deep	Tibialis anterior Extensor digitorum	First dorsal webspace	None
Superficial	Peroneus longus	Lateral lower leg, dorsal aspect of foot	None
Common	Tibialis anterior Extensor digitorum Peroneus longus	Lateral lower leg, dorsal aspect of foot	None
Tibial	Tibialis posterior Gastrocnemius	Plantar aspect of foot	Ankle

	Sensory and Motor	Motor	Sensory
Proximal	Symmetric Diabetic lumbosacral radiculoplexus neuropathy Asymmetric	Symmetric Spinal bulbar muscular atrophy, brachial amyotrophic diplegia, most myopathies Asymmetric	Symmetric Asymmetric
Proximal and Distal	Symmetric AIDP, CIDP Mononeuritis multiplex, MADSAM, HNPP Asymmetric	Symmetric SMA, PMA, infectious/ paraneoplastic MND, some myopathies (ie, FSHD), NMJ disease, pure motor CIDP ALS, leg amyotrophic diplegia, MMN, some myopathies (ie, IBM) Asymmetric	Symmetric Dorsal column dysfunction, CISP Sensory neuronopathy Asymmetric
Distal	Symmetric Distal symmetric polyneuropathy, CMT, DADS Asymmetric	Symmetric Distal hereditary motor neuropathies, distal myopathies (ie, DM1), MMN Hirayama disease Asymmetric	Symmetric Distal symmetric polyneuropathy Asymmetric

Localization/Disease	Pattern of Weakness	Distinguishing Clinical Features
Myopathy	Proximal predominant greater than distal predominant or multifocal	Variable
Neuromuscular junction		
Myasthenia gravis	Eyelids, eye movements, face, bulbar with or without limb involvement	Fatigability
Lambert-Eaton myasthenic syndrome	Lower extremity predominant	Hyporeflexia potentiated by exercise, dysautonomia
Motor neuron		
Amyotrophic lateral sclerosis	Limb or bulbar onset	Upper motor neuron signs, pseudobulbar affect, weight loss, frontotemporal dysfunction
Brachial amyotrophic diplegia	Bilateral proximal arms	No progression to other body segments for >12-18 months
Leg amyotrophic diplegia	One or both legs	No progression to other body segments for >12-24 months
Spinal muscular atrophy	Limb, bulbar and respiratory	Autosomal recessive inheritance pattern, infant or childhood onset
Spinal bulbar muscular atrophy (Kennedy disease)	Bulbar and proximal	Face and tongue fasciculations, gynecomastia, X-linked recessive inheritance pattern
Hirayama disease	Unilateral or bilateral hand and forearm	Young men, progresses for years then stabilizes, minipolymyoclonus, cold paresis
Infectious	Segmental or generalized	Acute or subacute onset, encephalitis, signs of systemic infection
Paraneoplastic	Segmental or generalized	Subacute onset, association with nonmotor paraneoplastic syndromes
Nerve		
Multifocal motor neuropathy	Distal upper extremity, weakness in the distribution of individual nerves; no bulbar or respiratory involvement	Weakness more than atrophy early in the course
Pure motor chronic inflammatory demyelinating polyradiculoneuropathy (CIDP)	Symmetric proximal and distal weakness; no bulbar or respiratory involvement	Reduced or absent reflexes

Localization	Examples of Disorders
Motor neuron	Acute flaccid paralysis, paraneoplastic
Spinal root	Disk herniation, varicella-zoster virus
Sensory neuron	Paraneoplastic, Sjögren syndrome, pyridoxine toxicity
Plexus	Idiopathic neuralgic amyotrophy, diabetic lumbosacral radiculoplexus neuropathy
Nerve	Direct nerve trauma, toxic (eg, chemotherapy or heavy metal-induced), vasculitic mononeuritis multiplex, acute inflammatory demyelinating polyradiculoneuropathy (AIDP) and variants
Neuromuscular junction	Botulism
Muscle	Inflammatory myopathies, rhabdomyolysis due to a metabolic myopathy, toxic myopathies

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Affected limb neuropathic pain suggest PNS > CNS lesion. Truncal sensory level, hyperreflexia, hemibody symptoms, arm extensor weakness & leg flexor weakness suggest CNS lesion

Neuralgic Amyotrophy may appear to localize to multiple nerves in the same limb rather than a specific part of the brachial plexus.

Injuries to nerve roots and mixed nerves, both of which contain both sensory & motor components, may present with pain or sensory symptoms without weakness.

CIDP progresses > 8 weeks after symptom onset & unlike AIDP, generally not with dysautonomia, cranial muscle weakness, or dyspnea.

Mononeuritis Multiplex affects named nerves, but not necessarily at common entrapment sites,

Mononeuritis Multiplex usually axonal, but **Multifocal Acquired Demyelinating Sensory & Motor Neuropathy (MADSMN)** & **Hereditary neuropathy with liability to pressure palsies (HNLPP)** are multifocal demyelinating neuropathies.

Distal symmetric polyneuropathy usually begins to involve distal upper extremities when lower extremity sensory symptoms progress up to the knees.

Hereditary Neuropathy red flags include young onset, family history, high arches & hammer toes

Spinal Bulbar Muscular Atrophy may cause face & tongue fasciculations, signs of androgen deficiency (e.g., gynecomastia) & slowly progressive bulbar & proximal weakness

Amyotrophic Lateral Sclerosis (ALS) - Pseudobulbar Affect (PBA) - emotional output dysregulation, not mood disorder, up to half cognitive / behavioral problems, overt dementia uncommon

Brachial amyotrophic diplegia & leg amyotrophic diplegia - painless flaccid weakness in one body segment, may or may not progress to other segments or cause UMN pathology

West Nile Virus & specific enterovirus systemic infections may cause polio-like **Acute Flaccid Paralysis (AFP)** with or without encephalitis

Extensor Digitorum Brevis (EDB) muscle may atrophy in sensorimotor distal symmetric polyneuropathy, but is often relatively spared in distal myopathies

ALS weakness tends to follow myotomal pattern, but **Multifocal Motor Neuropathy (MMN)** weakness may follow specific peripheral nerves distributions & may have significant weakness with little or no atrophy early in the course, suggesting motor axons still intact.

Hirayama Disease (Monomelic Amyotrophy) presents in young men with unilateral or bilateral hand weakness that progresses for years & then plateaus

Proprioception often involved early in disorders affecting dorsal columns & late in disorders affecting peripheral nerves

Sensory Neuronopathy may be idiopathic, with Sjögren's, paraneoplastic, HIV, or pyridoxine excess, initially appearing weak but normalizing when they look at the limb being tested

Distal Symmetric Polyneuropathy - screen for diabetes, lipid profile, vitamin B12, SPEP (serum protein electrophoresis) & immunofixation

Paraproteinemia may be with **CIDP**, common with **Distal Acquired Demyelinating Symmetric Neuropathy**, in most **POEMS (polyneuropathy, organomegaly, endocrinopathy, monoclonal plasma cell disorder & skin changes) syndrome**

Over Reliance on modest **CSF protein** elevations is a common reason for overdiagnosis of CIDP. **NCS / EMG** important for mononeuropathies, mononeuritis multiplex, demyelinating neuropathy, sensory neuropathy & motor neuron disease, but not sensitive for small fiber polyneuropathy or pure sensory radiculopathy

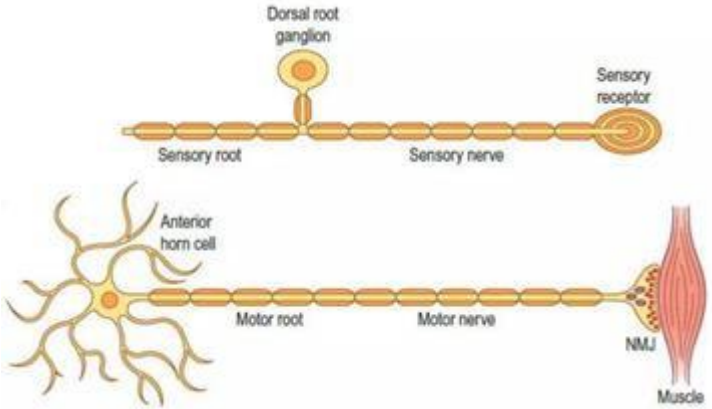
Spine **MRI** helpful in radiculopathy, CIDP & Hirayama disease, but may show degeneration of questionable significance in both symptomatic & asymptomatic individuals

Ultrasound is helpful with entrapment neuropathies & emerging indications in other neuromuscular disorders

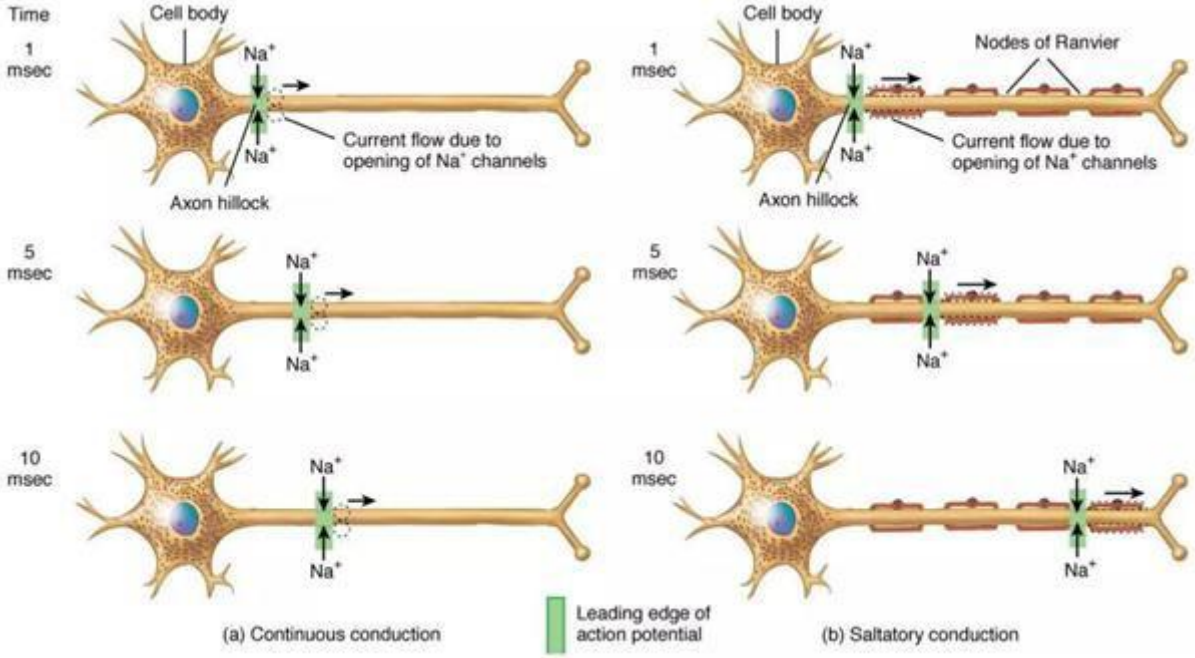
A **nerve biopsy** test of choice for **non-systemic vasculitic neuropathy**, but should only be performed in known **systemic vasculitis** if it will lead to a change in immunomodulatory treatment

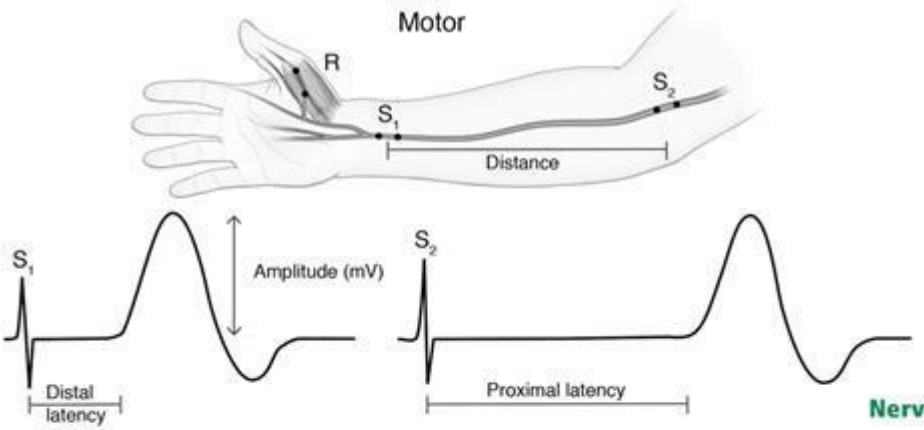
(Differential Pearls)

PNS = where Schwann cell is major supporting cell

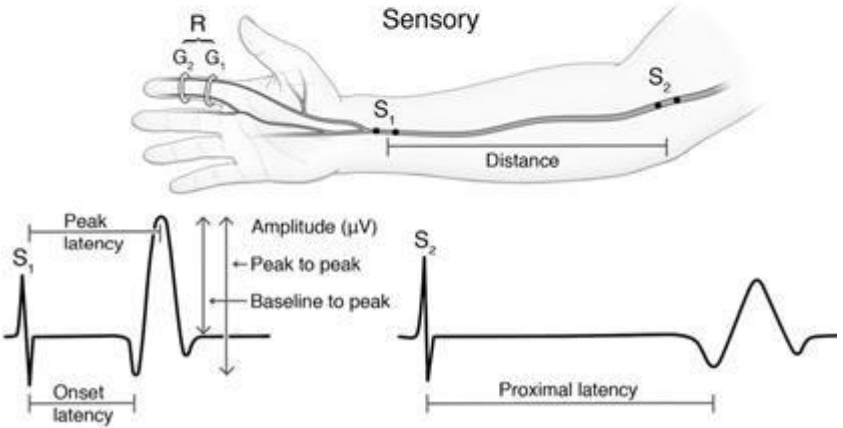


Saltatory v continuous conduction / propagation of NAP (Nerve Action Potential)

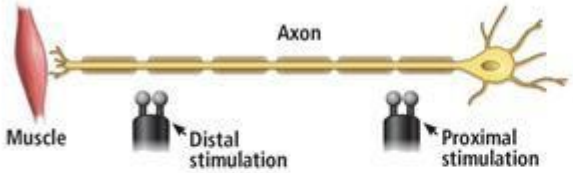


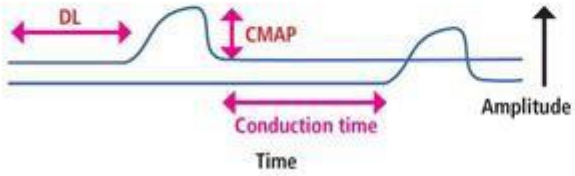






Nerve conduction studies: Principles



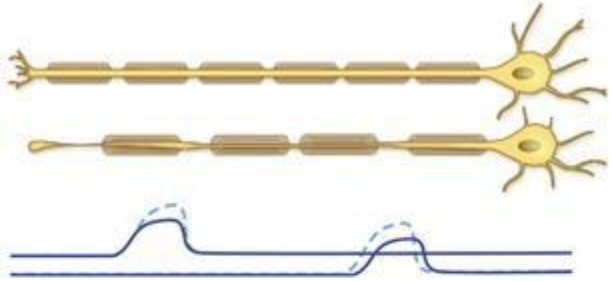


A motor nerve, composed of numerous axons (represented by a single neuron), is stimulated through the skin with a pulse of current administered through a stimulator, with enough current to depolarize all of the nerve's axons. Recording electrodes on the surface of the skin overlying the innervated muscle (not pictured) produce a tracing of electric potential over time, which represents the depolarization of all activated muscle cells.

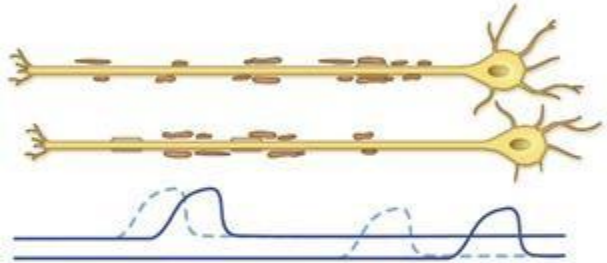
This deflection from the electric baseline is called the compound muscle action potential (CMAP). The time between stimulation at a distal site and the initial deflection of the CMAP is called the distal latency (DL), which is determined by the size and myelination of the motor nerve, as well as transmission across the neuromuscular junction and within the muscle itself. Motor nerves are often stimulated proximally as well, which allows for calculation of a conduction time and associated conduction velocity across a segment of the nerve. This parameter does not include the neuromuscular junction or intramuscular transmission, and represents purely nerve conduction within a nerve segment.

NCS

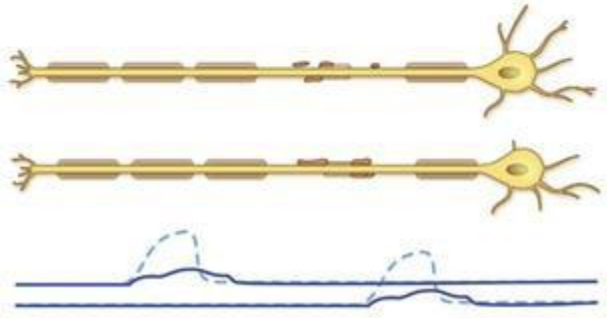
Nerve conduction studies: Abnormal patterns



Axon loss. When axons are lost, there are fewer excitable axons, and therefore fewer muscle cells are excited, resulting in a lower compound muscle action potential (CMAP) regardless of stimulation site. This can occur in peripheral neuropathy or motor neuron disease. The dashed tracings represent normal, solid tracings are abnormal.



Diffuse demyelination. For diffusely disrupted myelin, distal latency is prolonged, and the conduction velocity is slow, but the CMAP retains normal amplitude because all of the axons are still available to depolarize the same number of muscle cells. This may be seen in hereditary demyelinating neuropathies, such as Charcot-Marie-Tooth disease.



Focal demyelination. For focal demyelination over a portion of nerve, focal slowing occurs only over the affected segments. In addition, the conduction in some neurons is too slow to cross the area of focal demyelination. This is called conduction block, and results in a more than 50% reduction of CMAP amplitude when stimulating proximal to the lesion. Because focal demyelination typically affects different neurons to varying degrees, the action potentials arrive at the muscle at more variable times, leading to a "spreading" of the CMAP, known as temporal dispersion. This pattern can be seen in some types of compressive nerve injury (eg, ulnar neuropathy at the elbow) and diffuse acquired demyelinating polyneuropathies (eg, Guillain-Barré syndrome).

Condition	Latency	Conduction Velocity	Amplitude
Normal	Normal	Arm >45 m/s Leg >40 m/s	Normal
Axonal	Normal	Normal	Decreased
Demyelinating	**Increased**	**Decreased**	**Normal/decreased**

A healthy muscle should be silent at rest.

Abnormal activity *at rest* spontaneous single muscle fibers:

Fibrillations & Positive Sharp Waves:
often with denervation or myopathies

Fasciculations:
involuntary muscle twitches in chronic neurogenic conditions (e.g., ALS or radiculopathy)

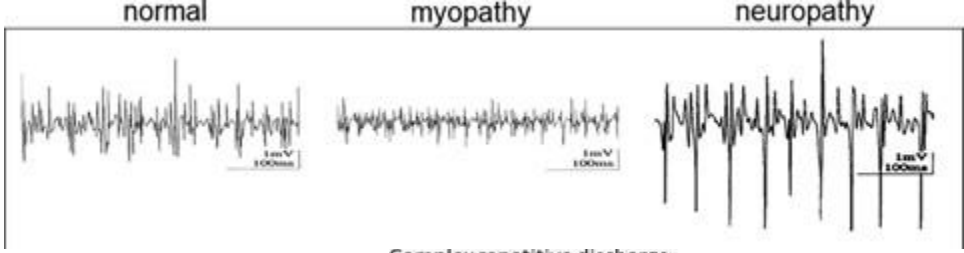
Abnormal Activity *During Contraction*: involves changes to the motor unit potentials

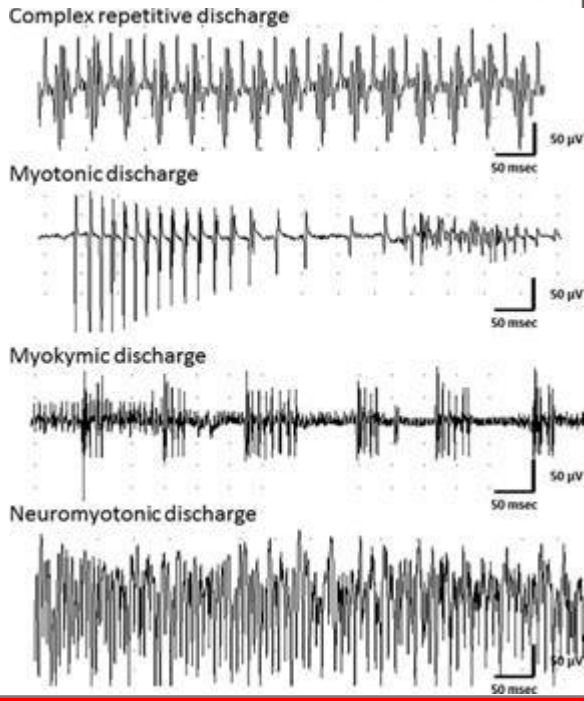
Waveform changes:
unusually small, large, or irregularly shaped waves

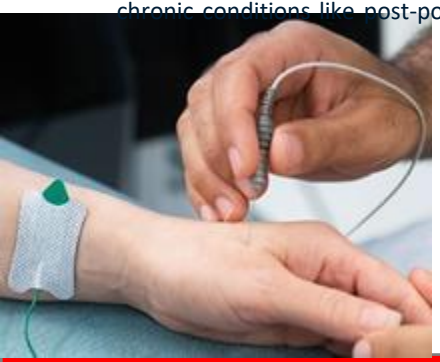
Reduced Interference Pattern:
healthy muscle - many overlapping potentials during maximal contraction

Large Polyphasic Motor Units:
complex, multi-peaked waveforms -
denervation & reinnervation of muscle fibers by surviving nerve sprouts

Giant Motor Units:
extremely large & wide motor units, long time to develop,
chronic conditions like post-polio syndrome or chronic neuropathies

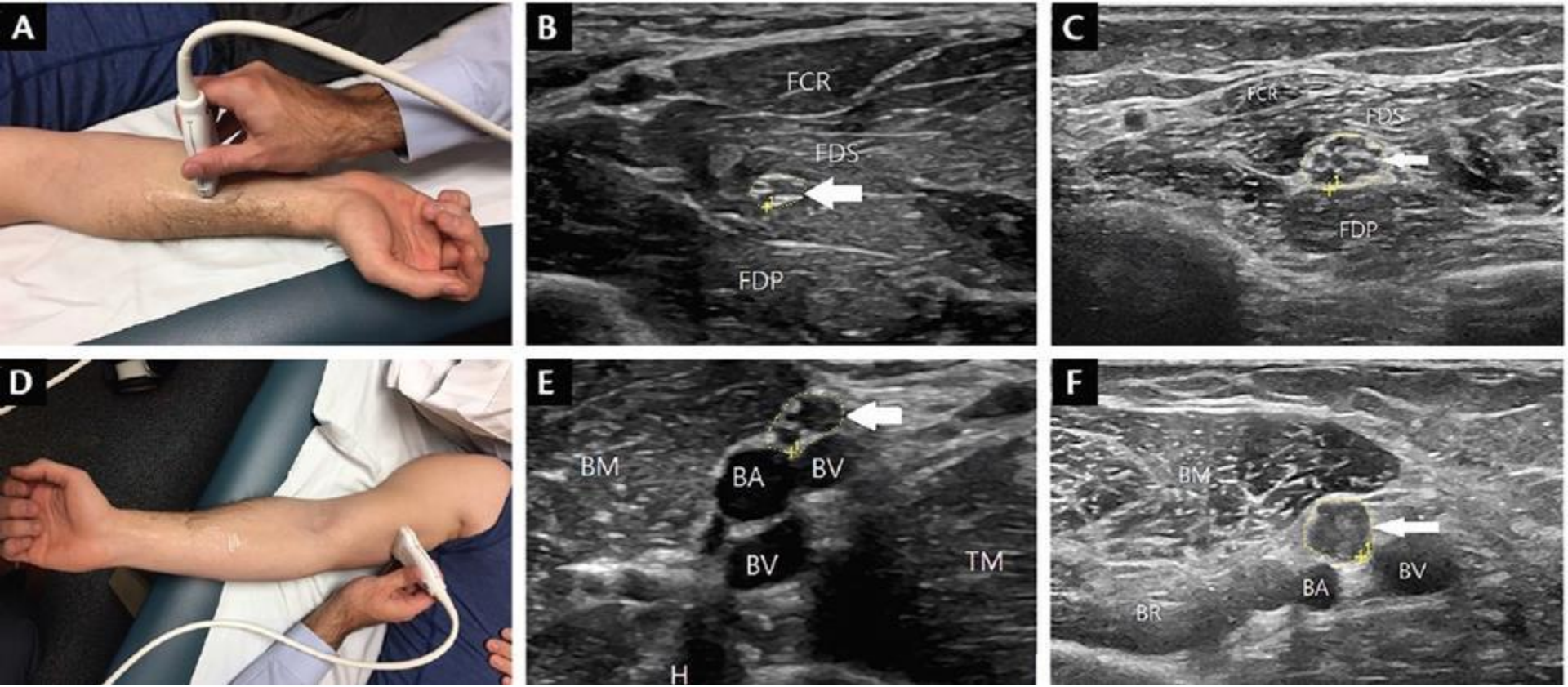








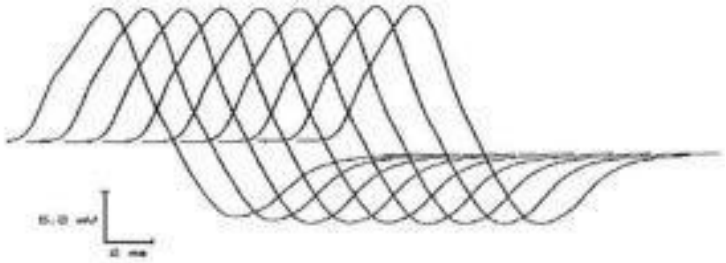
Needle EMG



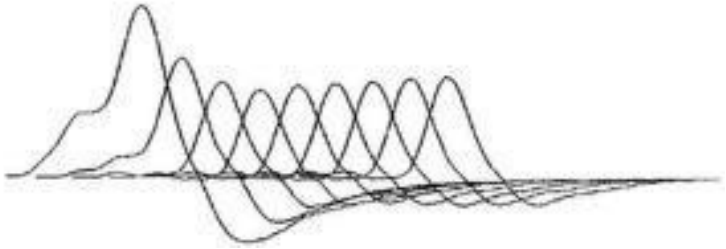
Differentiates demyelinating inflammatory neuropathies (ie, CIDP, MADSAM, MMN), from axonal neuropathies based on enlargement of the proximal median nerve & brachial plexus (high sensitivity 95% & specificity 98%)

Repetitive Nerve Stimulation

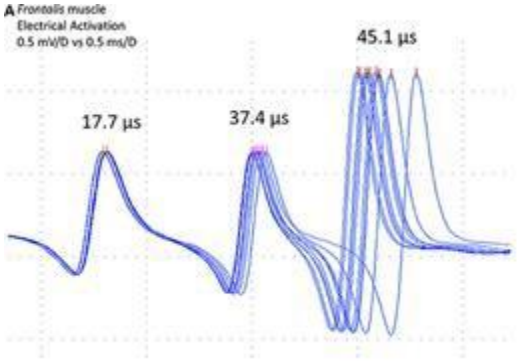
Normal

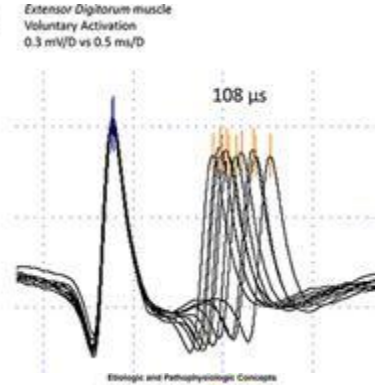


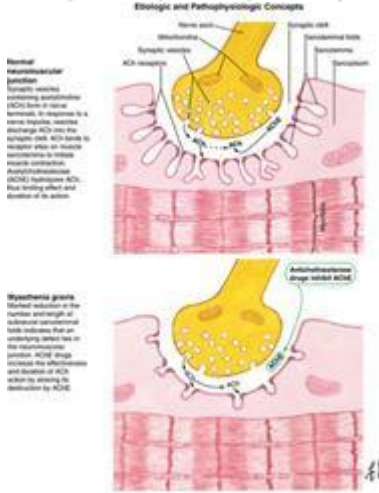
MG



Single Fiber EMG Jitter







Myasthenia gravis (MG) *decremental* response high-rate RNS

autoimmune disorder affecting neuromuscular junction (NMJ) (Abs against postsynaptic Ach receptors), fatigable muscle weakness (worsens with activity), often begins ocular => generalized, thymoma

Lambert-Eaton Myasthenic Syndrome (LEMS) *incremental* response at high-rate RNS

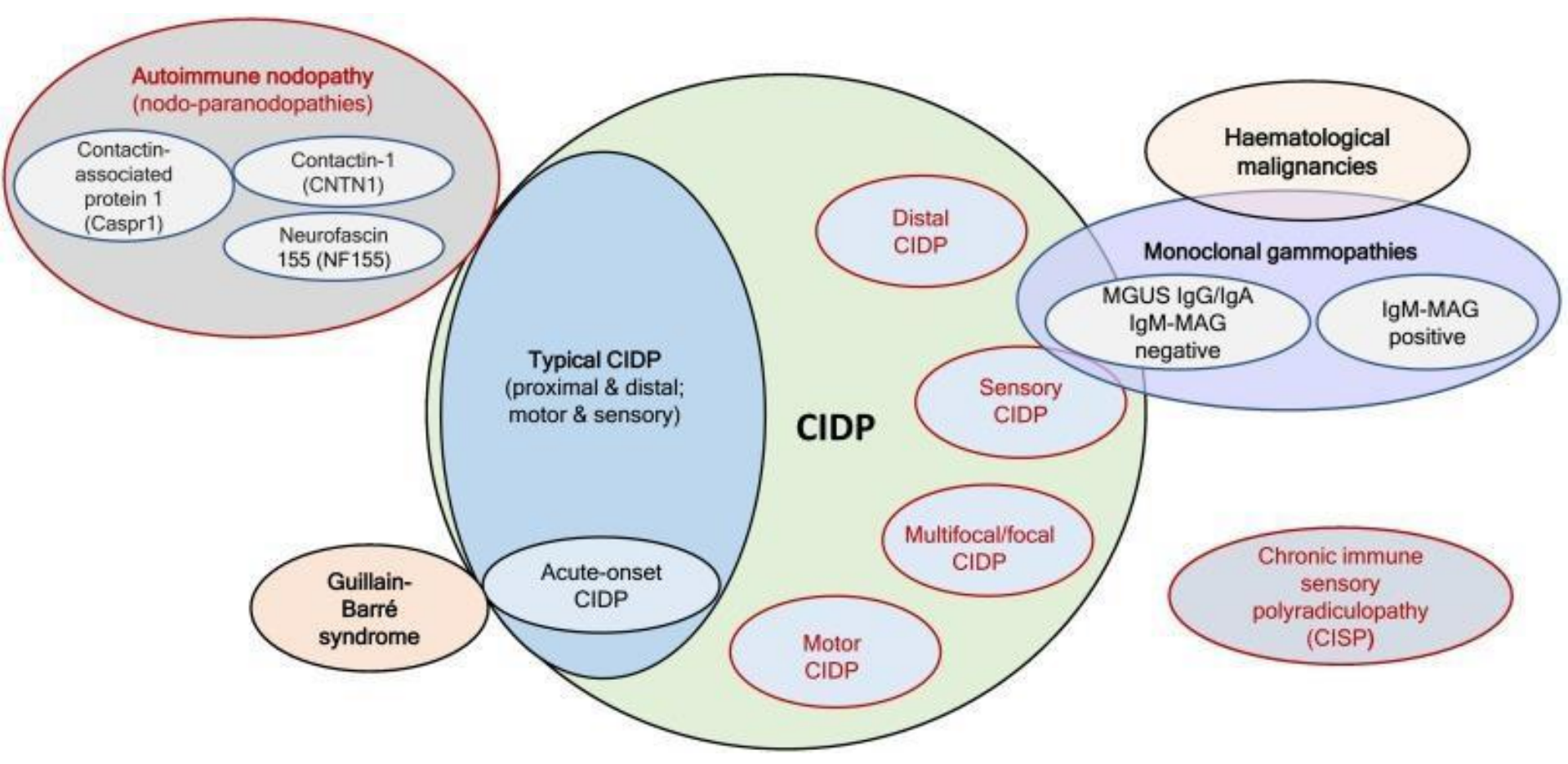
paraneoplastic (SCLC) autoimmune disorder - Abs attack presynaptic Voltage-Gated Calcium Channels (VGCC), disrupting neurotransmitter release, limb weakness briefly improves with exercise, ANS involvement unlike MG

RNS v Single Fiber EMG

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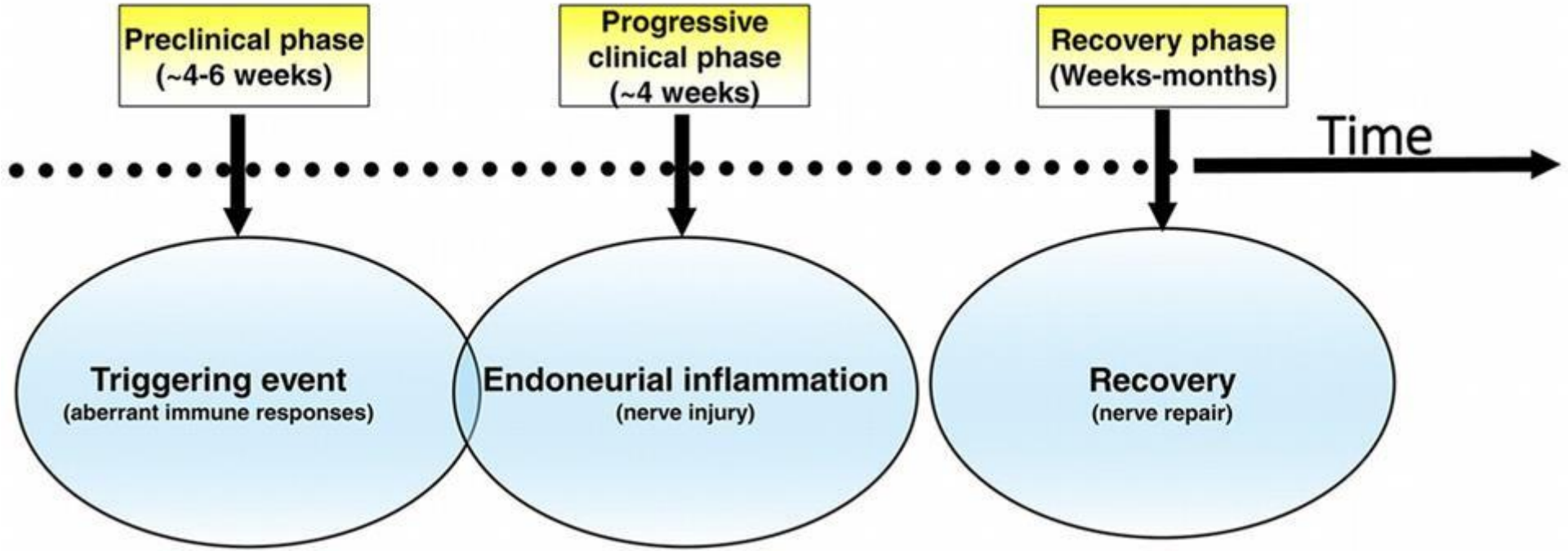
Shifting Gears



CIDP Spectrum

Features Required for Guillain-Barré Syndrome Diagnosis	Features Supportive of Diagnosis	Features Casting Doubt on Diagnosis	Features That Rule Out Diagnosis
Progressive muscle weakness of more than one limb	Progression of weakness for 2-4 weeks	Marked persistent asymmetry of weakness	History of hexacarbon abuse
Areflexia or hyporeflexia	Symmetric involvement Mild sensory symptoms or signs Cranial nerve involvement Recovery begins 2-4 weeks after nadir Autonomic dysfunction Absence of fever Elevated CSF protein	Persistent bladder or bowel dysfunction Severe bladder or bowel dysfunction at onset >50 Leukocytes/mm^3 in CSF Presence of polymorphonuclear leukocytes in CSF Sharp sensory level Central nervous system signs except in Miller Fisher syndrome-Bickerstaff brainstem encephalitis spectrum disorders	Acute intermittent porphyria Recent diphtheritic infection Poliomyelitis, botulism, toxic neuropathy, functional paralysis

GBS NINDS Criteria



Monophasic GBS

Acute Radiculo/neuropathy	Diagnostic Clues
Inflammatory/immune	
Acute-onset chronic inflammatory demyelinating polyradiculoneuropathy (CIDP)	Progression >4 weeks or three treatment-related fluctuations in first 8 weeks
Vasculitic neuropathy	Frequently asymmetric (can be confluent), predominantly axonal
Sensory ganglionitis	Diffuse sensory loss, areflexia, pseudoathetosis, pseudoweakness (may improve with visual compensation), posterior column degeneration on MRI with latency of approximately 10 weeks
Intensive care unit setting	
Critical illness neuropathy	In the setting of sepsis or multiorgan failure, often ventilator dependence, concomitant myopathy
Metabolic	
Diabetes	Frequently asymmetric (lumbosacral plexus), weight loss, history of diabetes mellitus
Porphyria	Axonal physiology, predominantly motor, psychiatric features, abdominal pain, positive metabolic screen
Nutritional	
Thiamine deficiency	Often history of gastric bypass surgery, alcohol use disorder, vomiting, acute weight loss, other B vitamins can be low, axonal
Infections	
Human immunodeficiency virus (HIV)	Early in infection, positive testing, CSF pleocytosis
Herpesviruses, including cytomegalovirus	Often in setting of HIV, mixed myelopathic and radicular features, CSF pleocytosis (polymorphonuclear leukocytes with cytomegalovirus), polymerase chain reaction (PCR)
Lyme disease	Radiculoneuritis, asymmetric, endemic area, tick bite, erythema migrans, CSF pleocytosis, predominantly axonal
Toxic	
Organophosphates	Exposure to insecticides, occupation in plastic and petroleum industries, axonal
Hexacarbons	Occupational exposure or inhalant abuse of glues, lacquers, cleaning solvents, axonal; secondary demyelinating changes including multifocal motor conduction block can be seen

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Guillain-Barré syndrome (GBS) spectrum of acute neuropathic disorders, muscle weakness cardinal manifestation in the majority, most common cause of acute flaccid paralysis (AFP)

Paralytic GBS NINDS criteria: symmetric flaccid weakness, decreased deep tendon reflexes, exclusion of alternative causes.

Acute Inflammatory Demyelinating Polyradiculoneuropathy (AIDP) 1st symptoms often sensory, primarily motor symmetric proximal & distal weakness, classically ascending, also proximal

25-30% will require intubation due to respiratory & pharyngeal muscle weakness

Miller Fisher Syndrome - related disorders clinical continuum: **Bickerstaff Brainstem Encephalitis** (altered level of consciousness or hyperreflexia, external ophthalmoplegia, ataxia) <=>

Miller Fisher Syndrome - most common minor subtype, triad ophthalmoplegia, ataxia, areflexia

GBS common residual symptoms - fatigue, pain, paresthesia, weakness

NCS / EMG acute neuropathic process, demyelinating vs axonal variants, often relatively normal early, F waves

Acute Motor Axonal Neuropathy (AMAN) reversible conduction failure at nodes of Ranvier - partial motor nerve conduction block w/o temporal dispersion,

w/o other demyelinating features reduced conduction velocity, prolonged minimal F-wave or distal motor latencies

GBS CSF typically **albuminocytologic dissociation**, mild pleocytosis < 50 cells/mm3 up to 10-15%, > 50 cells/mm3 suggests alternate diagnosis

GBS prognostic models per clinical parameters, Medical Research Council sum score, reliably predict need for mechanical ventilation in 1st week & functional outcomes at 4 weeks to 6 months

AIDP, AMAN & acute motor-sensory axonal neuropathy (AMSAN) activated innate immune system - complement activation & IgG Fc receptors (FcγRs) upregulation - therapeutic targets

GBS environmental trigger in genetically susceptible hosts: ***Campylobacter jejuni*** most common, particularly axonal forms, ~prevalence 30%, low risk of GBS with *C. jejuni* infection < 2 / 10K

Rare non-infectious triggers: trauma, vaccinations, immunosuppression, pregnancy

Risk of GBS following influenza infection much higher than following vaccination, in either case should not receive the same vaccine again.

Post-infectious molecular mimicry - predominant pathophysiologic mechanism in Miller Fisher syndrome & axonal variants & AIDP following *Mycoplasma pneumoniae* infection

Supportive & intensive care - acute management of monophasic GBS; immunotherapy with **plasma exchange (Plex) or IV immunoglobulin (IVIg)** hastens recovery - are equally efficacious with no added benefit with combination therapy per RCTs, but 40-50% did not have a clinical response that met primary endpoint), emphasizing need for new therapies

Biologics targeting complement cascade - clinical trial various stages - neonatal Fc receptor (FcRn) inhibitors (reduce IgG autoantibody burden) & FcγR modulators - advanced development

GBS Key Points

Chronic Inflammatory Demyelinating Polyradiculoneuropathy Variant	Clinical Features	Percentage of Chronic Inflammatory Demyelinating Polyradiculoneuropathy Cases	Treatment Response	Notes
Typical chronic inflammatory demyelinating polyradiculoneuropathy (CIDP)	Symmetric proximal and distal weakness, sensory loss, areflexia	50%[4]	Immunoglobulins, corticosteroids, plasma exchange	Clinical nadir reached after 8 weeks
Acute CIDP	Symmetric proximal and distal weakness, sensory loss, areflexia	Up to 18%[6,7]	Immunoglobulins, corticosteroids, plasma exchange	Clinical nadir reached before 8 weeks; may mimic Guillain-Barré syndrome
Sensory CIDP	Symmetric sensory-predominant presentation, sensory ataxia, generalized hyporeflexia/areflexia	5-15%[5,8]	Immunoglobulins, corticosteroids, plasma exchange	Distinguished from idiopathic sensory polyneuropathy by early ataxia, younger age, early upper extremity symptoms
Chronic immune sensory polyradiculopathy (CISP)	Sensory ataxia	Unknown, very rare	Immunoglobulins, corticosteroids	Normal nerve conduction studies; diagnosis relies on somatosensory evoked potentials, CSF analysis, and MRI abnormalities of lumbar roots
Motor CIDP	Symmetric proximal and distal motor deficits	4-10%[4,8,9]	Immunoglobulins; avoid corticosteroids	May have electrophysiologic evidence of sensory involvement
Multifocal acquired demyelinating sensory and motor neuropathy (MADSAM)	Asymmetric distal more than proximal sensory and motor deficits; affects upper more than lower extremities	8-15%[5,9]	Immunoglobulins, corticosteroids, plasma exchange	May evolve into typical CIDP
Distal acquired demyelinating and symmetric (DADS) neuropathy	Symmetric distal more than proximal, sensory more than motor deficits; affects lower more than upper extremities	2-10%[9,10]	With myelin-associated glycoprotein (MAG) antibodies, treatment ineffective although rituximab response reported in some series; without MAG antibodies, response to immunoglobulins, corticosteroids, plasma exchange	May be associated with an IgM monoclonal gammopathy; may have MAG antibodies; considered DADS-CIDP in the absence of monoclonal gammopathy and MAG antibodies

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CIDP Electrodiagnostic Criteria [a,b] EFNS/PNS

Definite (at least one of the following):

- Motor distal latency prolongation ≥50% above upper limit of normal values (ULN) in two nerves (excluding median neuropathy at the wrist from carpal tunnel syndrome), *OR*
- Reduction of motor conduction velocity ≥30% below lower limit of normal values (LLN) in two nerves, *OR*
- Prolongation of F-wave latency ≥30% above ULN in two nerves (≥50% if amplitude of distal negative peak compound muscle action potential [CMAP] <80% of LLN values), *OR*
- Absence of F waves in two nerves if the nerves have distal negative peak CMAP amplitudes ≥20% of LLN + ≥1 other demyelinating parameter[c] in ≥1 other nerve, *OR*
- Partial motor conduction block: ≥50% amplitude reduction of the proximal negative peak CMAP relative to distal, if distal negative peak CMAP ≥20% of LLN, in two nerves, or in one nerve + ≥1 other demyelinating parameter[c] in ≥1 other nerve, *OR*
- Abnormal temporal dispersion (>30% duration increase between the proximal and distal negative peak CMAP) in ≥2 nerves, *OR*
- Distal CMAP duration (interval between onset of the first negative peak and return to baseline of the last negative peak) increase in ≥1 nerve (median ≥6.6 ms, ulnar ≥6.7 ms, peroneal ≥7.6 ms, tibial ≥8.8 ms)[34] + ≥1 other demyelinating parameter[c] in ≥1 other nerve

Probable

- ≥30% amplitude reduction of the proximal negative peak CMAP relative to distal, excluding the posterior tibial nerve, if distal negative peak CMAP ≥20% of LLN, in two nerves, or in one nerve + ≥1 other demyelinating parameter[c] in ≥1 other nerve

Possible

- As in definite but in only one nerve

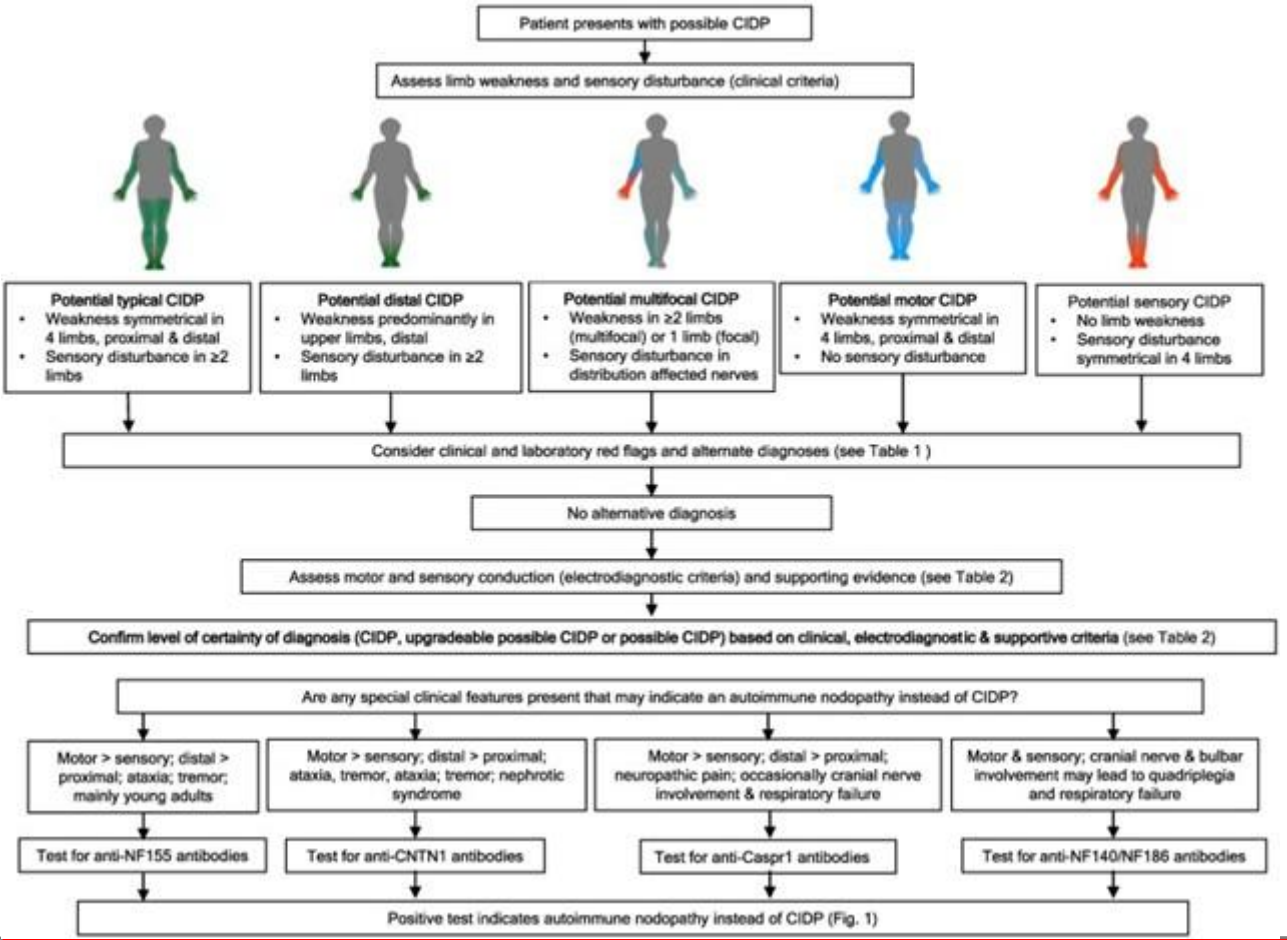
[a] Modified with permission from Van den Bergh PYK, et al, Eur J Neurol.[33] © 2010 The Authors. Journal compilation © 2010 EFNS and Peripheral Nerve Society.

[b] To apply these criteria, the median, ulnar (stimulated below the elbow), fibular (peroneal) (stimulate below the fibular head), and tibial nerves on one side are tested. If criteria are not fulfilled, the same nerves are tested at the other side and/or the ulnar and median nerves are stimulated bilaterally at the axilla and at Erb's point. Motor conduction block is not considered in the ulnar nerve across the elbow, and at least 50% amplitude reduction between Erb's point and the wrist is required for probable conduction block. Temperatures should be maintained to at least 33°C (91.4°F) at the palm and 30°C (86°F) at the external malleolus (good practice points).

[c] Any nerve meeting any of the criteria (1 through 7).

CIDP Misdiagnosis from Electrophysiologic Pitfalls

- Equivocal degrees of conduction velocity slowing in the axonal polyneuropathies
- Prolonged fibular (peroneal) nerve distal motor latency in the setting of reduced amplitude
- Mild conduction velocity slowing observed in motor neuron disease
- Conduction velocity slowing limited to sites of compression
- Slowing of sensory responses and amplitude-independent slowing in the context of diabetes
- Conduction velocity slowing and prolongation of distal latencies due to cold limb temperatures



CIDP Variants

Chronic Inflammatory Demyelinating Polyradiculoneuropathy (CIDP)

~50% typical onset symmetric proximal & distal weakness, length-dependent loss of large fiber sensation, areflexia.

Up to 18% acute onset mimics **Guillain-Barré Syndrome (GBS)**

CIDP **differentiated** from GBS by protracted course, absent autonomic dysfunction & absent respiratory impairment in most

Screen all suspected CIDP for **monoclonal gammopathy** (SPEP / IF)

CSF **albuminocytologic dissociation** expected in CIDP. Leukocytosis raises suspicion for other causes - neurosarcoidosis, HIV or carcinomatous meningitis

Sural sparing pattern - CIDP electrophysiologic hallmark, often found in addition to other acquired demyelinating features

Suspected CIDP don't need **nerve biopsy** if NCS & clinical features are consistent with **acquired demyelinating polyradiculoneuropathy**

> 15 sets CIDP published diagnostic criteria - most widely accepted *European Federation of Neurological Sciences/Peripheral Nerve Society* criteria

1st line treatments **immunoglobulins (IV & SQ)**, corticosteroids, plasma exchange (2nd or 3rd line - long-term venous access & limited outpatient facilities)

IVIg can be successfully discontinued w/o relapse in ~50% of patients

Multifocal Motor Neuropathy (MMN) may mimic **Amyotrophic Lateral Sclerosis (ALS)** painless progressive weakness - but no UMN signs & motor NCS conduction block

Distal Acquired Demyelinating Symmetric (DADS) Neuropathy associated with **IgM monoclonal gammopathy** & **Myelin-Associated Glycoprotein antibodies (anti-MAG)** DADS often refractory to treatment

IgG4 isotype autoantibodies directed toward paranodal & nodal antigens ~ 10% "CIDP"; unique presentations, often IVIg refractory - ? rituximab responsive

Autoimmune Nodopathies no longer considered seropositive CIDP - stand alone

CIDP Key Points

Chronic Inflammatory Demyelinating Polyneuropathy

Rare immune-mediated, prevalence 2.81/100K, incidence 0.33/100K, ~420K globally, more common older men peak 70-79 yo
Chronic Acquired Demyelinating Polyneuropathies: CIDP & variants, POEMS, Anti-MAG Neuropathy (DADS), Autoimmune Nodopathies, Multifocal Motor Neuropathy (MMN)
CIDP & variants: heterogeneous pathogenic mechanisms: (Distal & Sensory CIDP), Multifocal CIDP likely T cells & macrophages attack focal portions; may damage distal & proximal nerves where blood-nerve barrier is deficient, likely Ab & humoral factor mediated, also complement
Progressive weakness, sensory symptoms, lack of bulbar symptoms
Typical CIDP: ~ 50%, symmetrical proximal & distal weakness, hyporeflexia/areflexia, distal large fiber sensory loss
Variants/Atypical: Distal , Focal / Multifocal, Sensory predominant, Motor predominant, Autoimmune Nodopathies (now stands alone , no longer sero+ CIDP)
Misdiagnosis common ~ 50%
Always check HgbA1c, SPEP (Serum Protein Electrophoresis) / serum & urine IF (immunofixation), hepatic panel, genetic CMT, B12
Paraproteinemia - plasma cell malignancy excluded then MGUS (Monoclonal Gammopathy Unclear Significance)

Kelly Gwathmey, MD, Virginia Commonwealth University, AAN Fall Conference 2024

Antibody Target	Clinical Features	Percentage of Chronic Inflammatory Demyelinating Polyradiculoneuropathy Cases[a]	Target	Treatment Response
Neurofascin 155 (NF155)	Subacute onset, symmetric motor more than sensory, sensory ataxia, tremor (also of head/voice), distal-predominant; cranial neuropathies reported	5–10%	Paranodal transmembrane cell adhesion molecule located on Schwann cell membrane	Good response to rituximab and plasma exchange; partial response to corticosteroids; IV immunoglobulin (IVIg) refractory
Neurofascin 140 (NF140)/ neurofascin 186 (NF186)	Subacute onset, symmetric sensory and motor, sensory ataxia, cranial neuropathies; nephrotic syndrome reported	2%	Axonal membrane protein involved in clustering of sodium channels located at the nodes of Ranvier; interacts with gliomedin and neuronal cell molecules	Good response to rituximab; partial response to IVIg and corticosteroids
Contactin-1 (CNTN1)	Subacute onset, symmetric motor more than sensory, sensory ataxia; nephrotic syndrome reported	5%	Paranodal, axonal cell adhesion molecule	Good response to rituximab; partial response to corticosteroids; IVIg refractory
Contactin-associated protein 1 (CASPR1)	Subacute onset, symmetric distal-predominant, severe neuropathic pain	<1%	Paranodal, axonal cell adhesion molecule	Good response to rituximab; IVIg refractory

Nodo-Paranodopathies

rdare

“If you get confused,

listen to the music play”



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